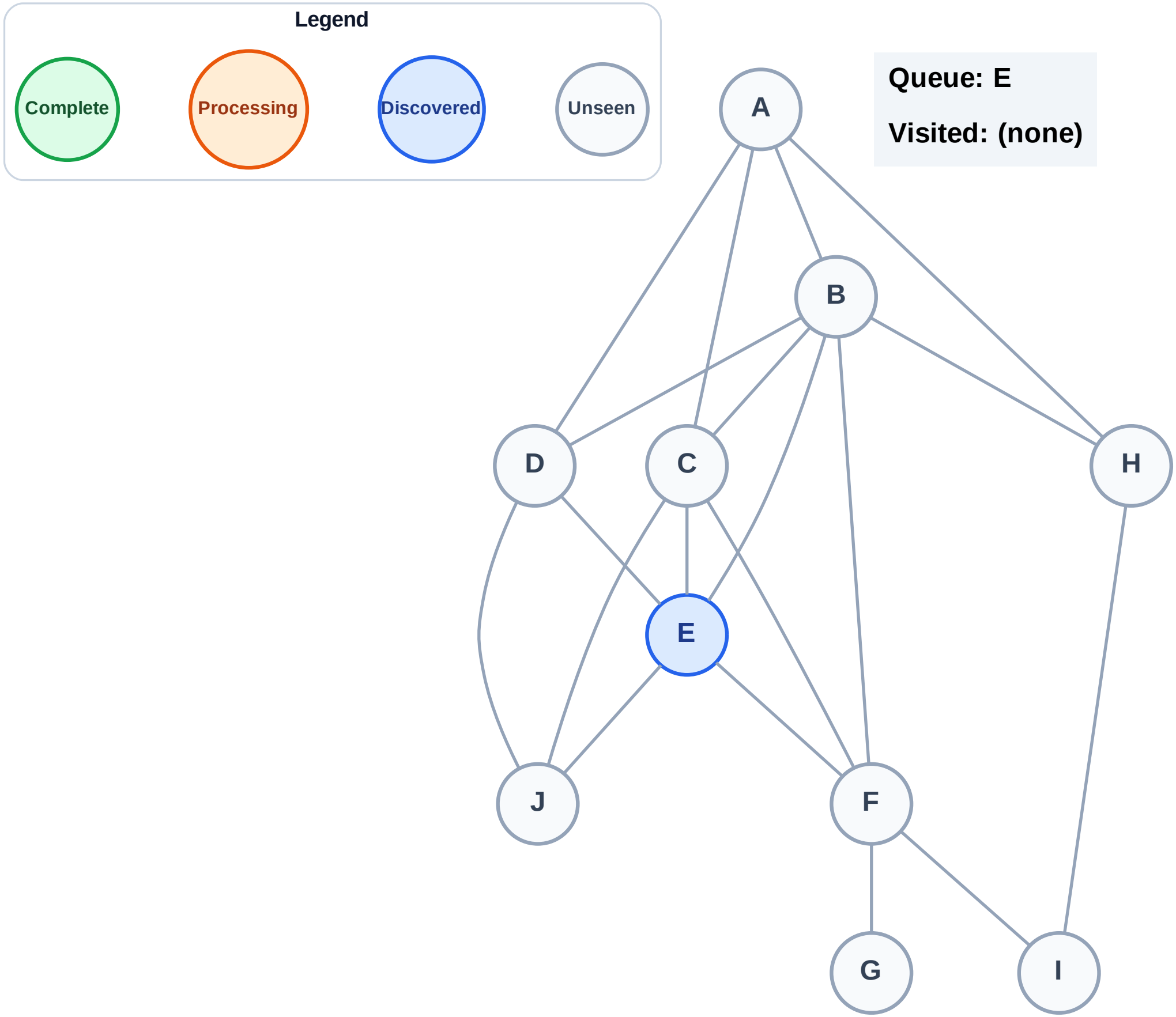
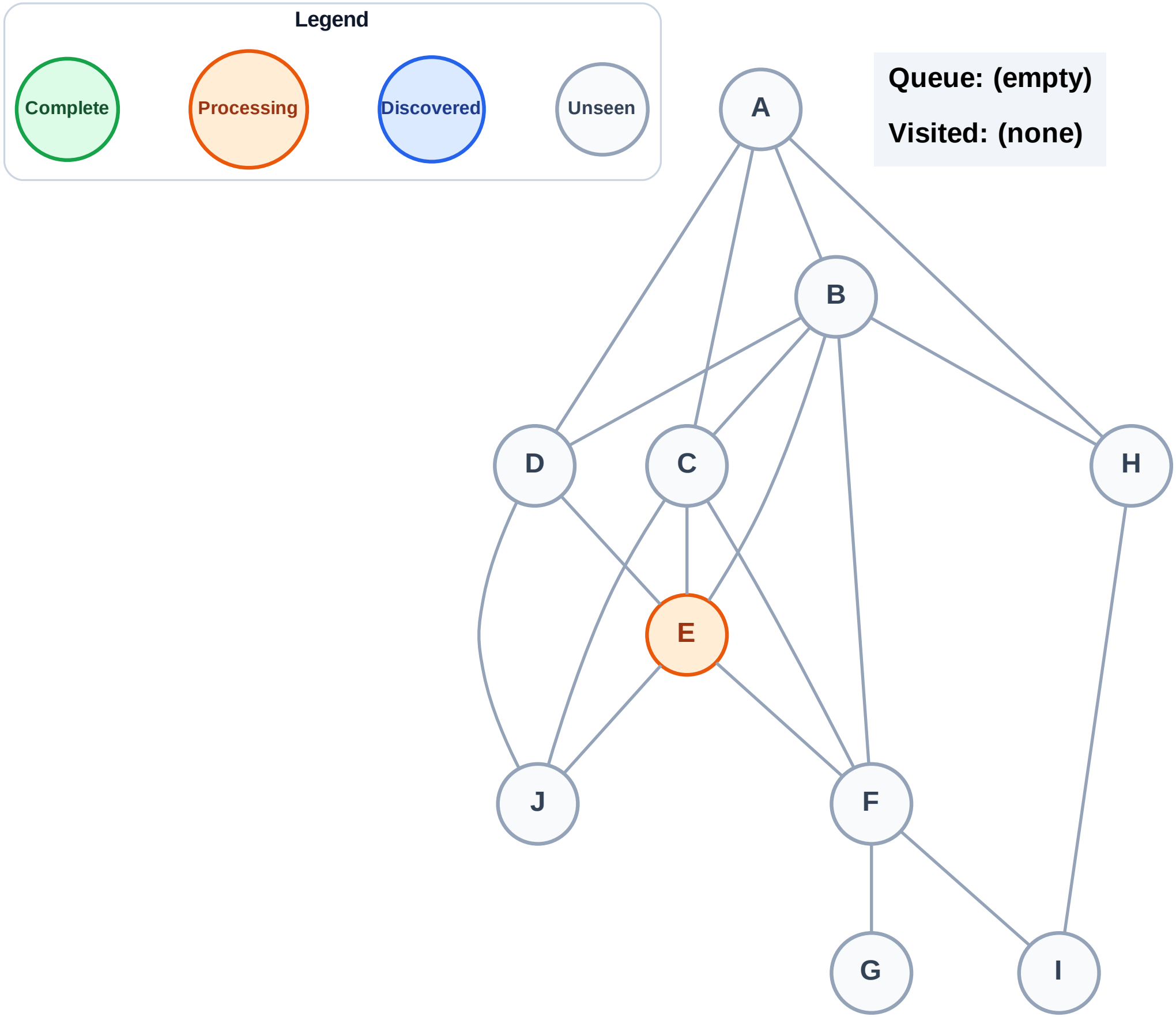
BFS Traversal

Step 1: Start at E



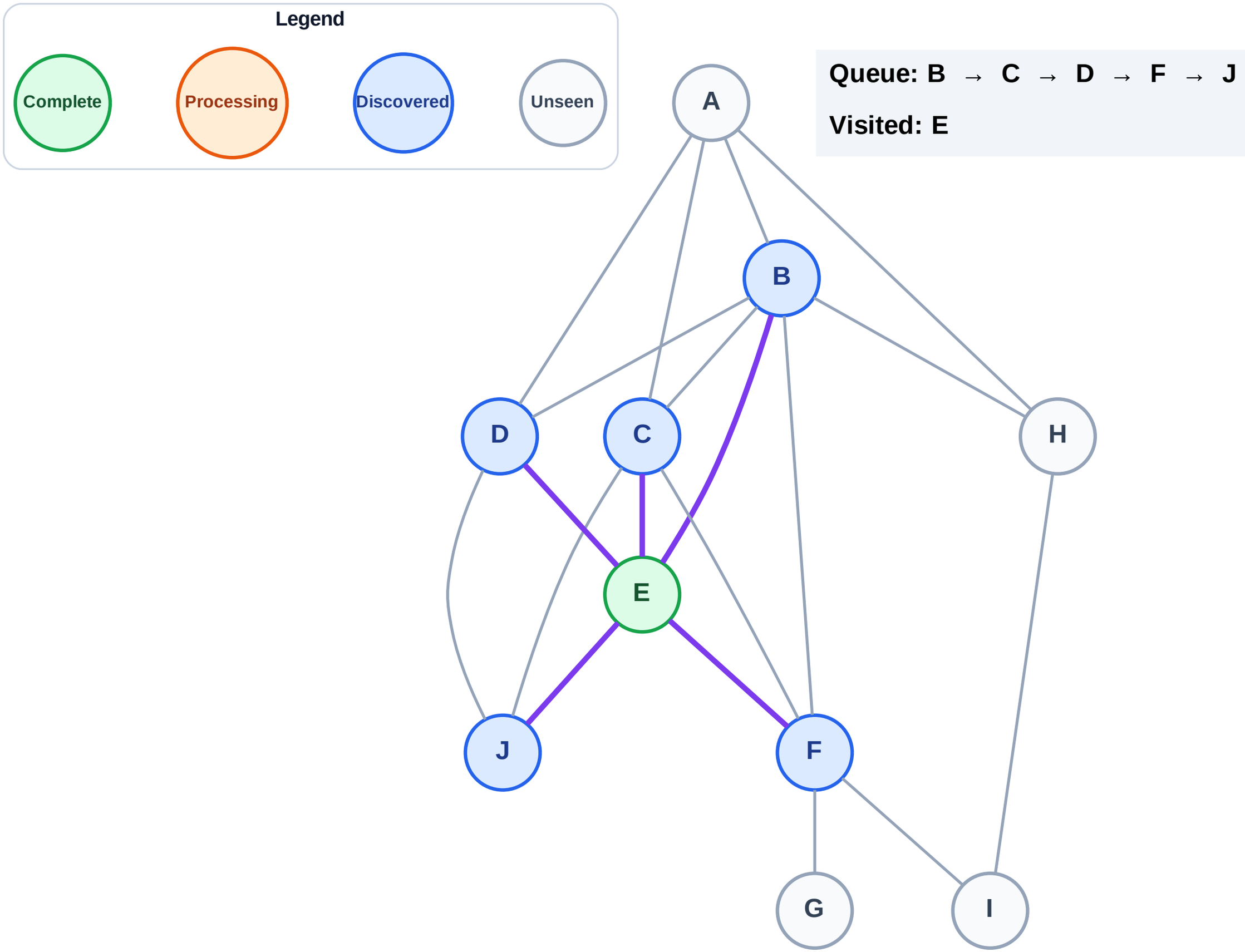
BFS Traversal

Step 2: Process node E



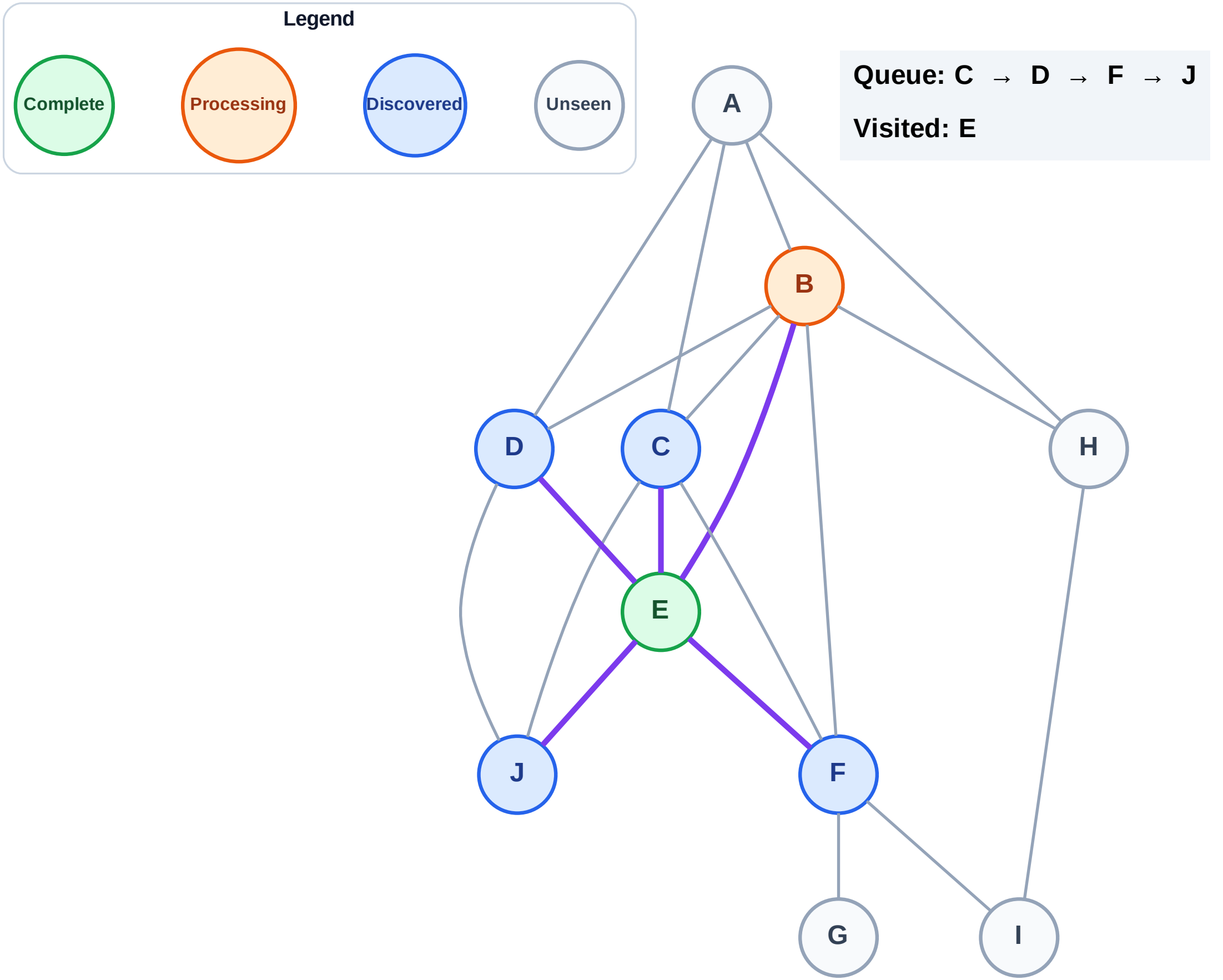
BFS Traversal

Step 3: Discover B, C, D, F, J from E



BFS Traversal

Step 4: Process node B

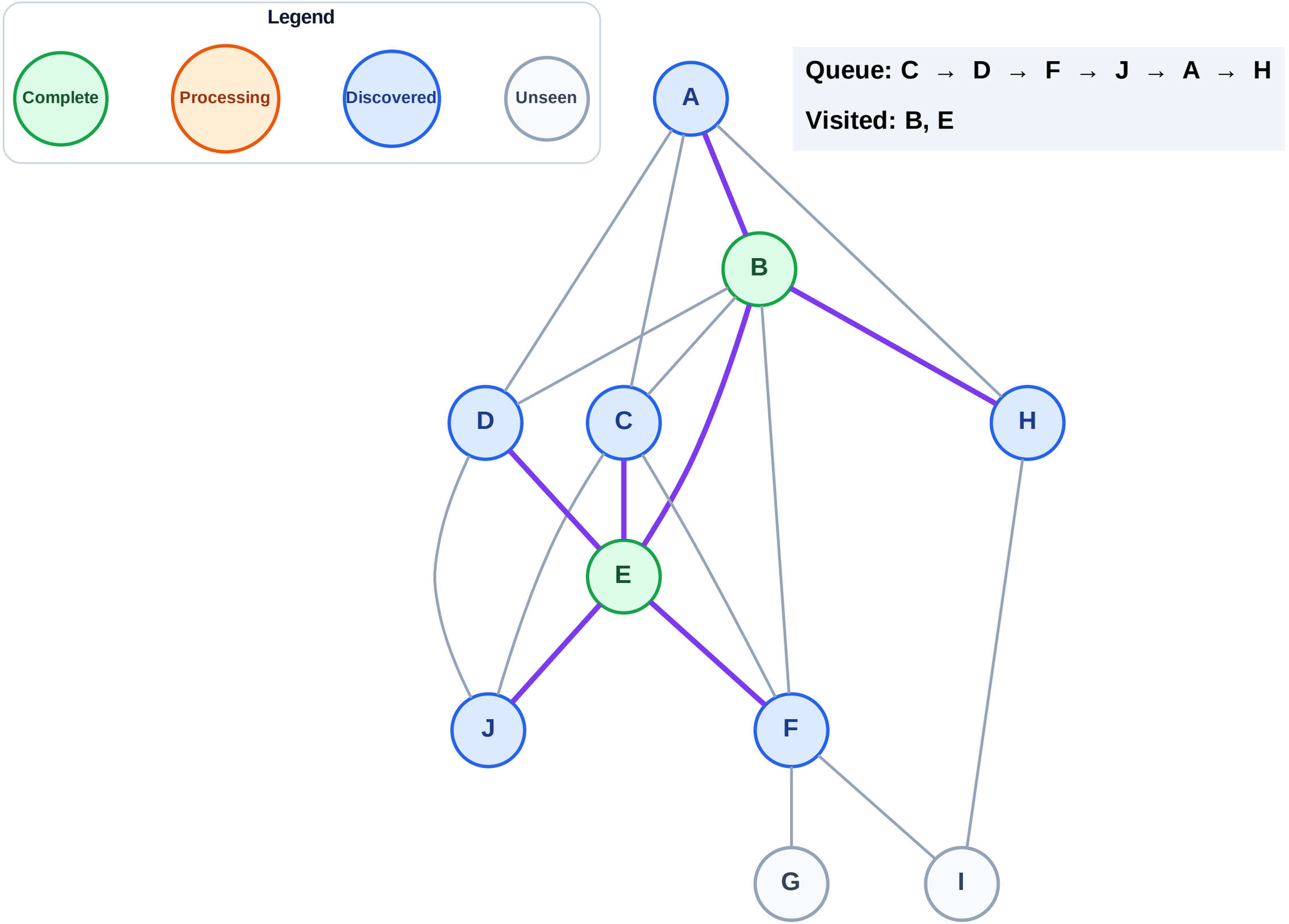


Queue: C → D → F → J

Visited: E

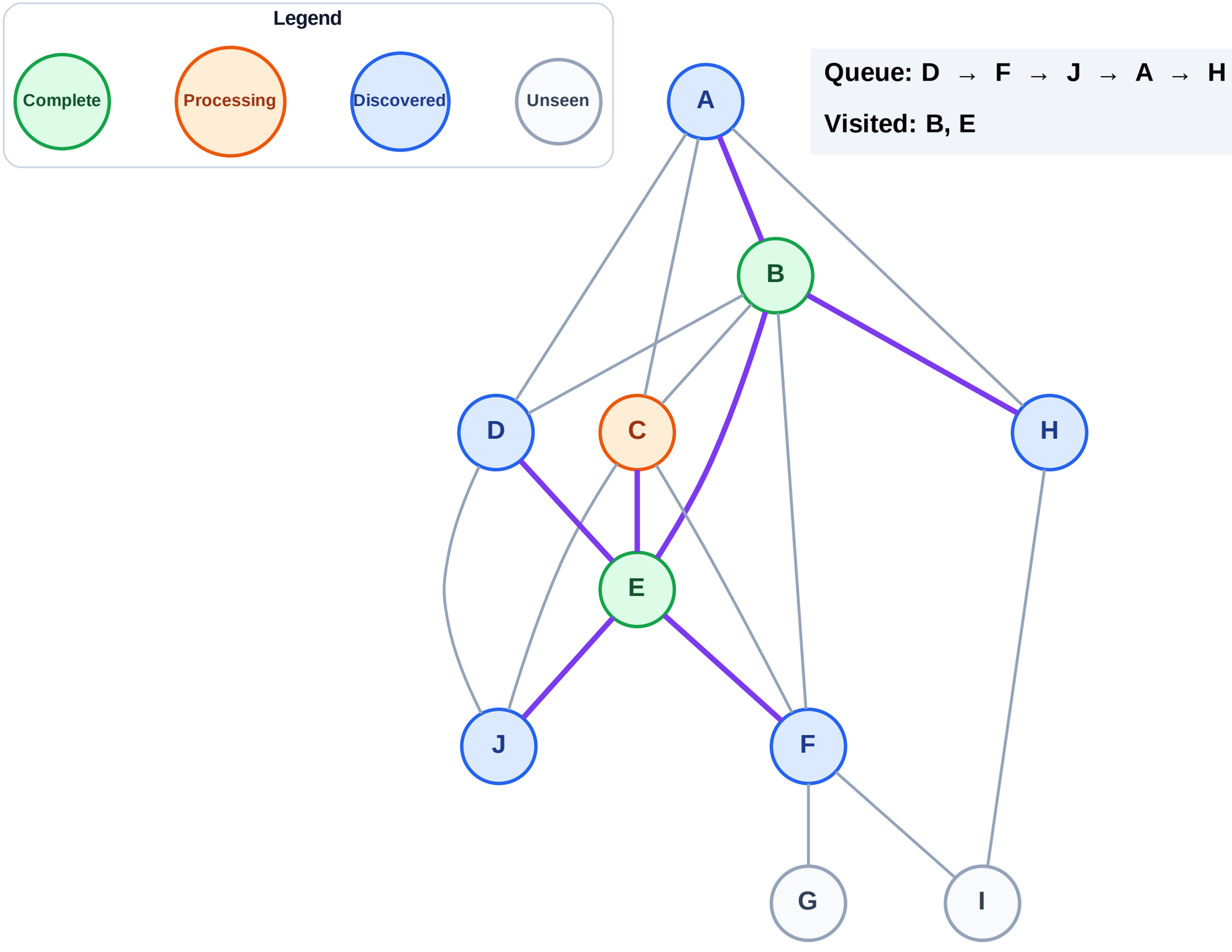
BFS Traversal

Step 5: Discover A, H from B



BFS Traversal

Step 6: Process node C

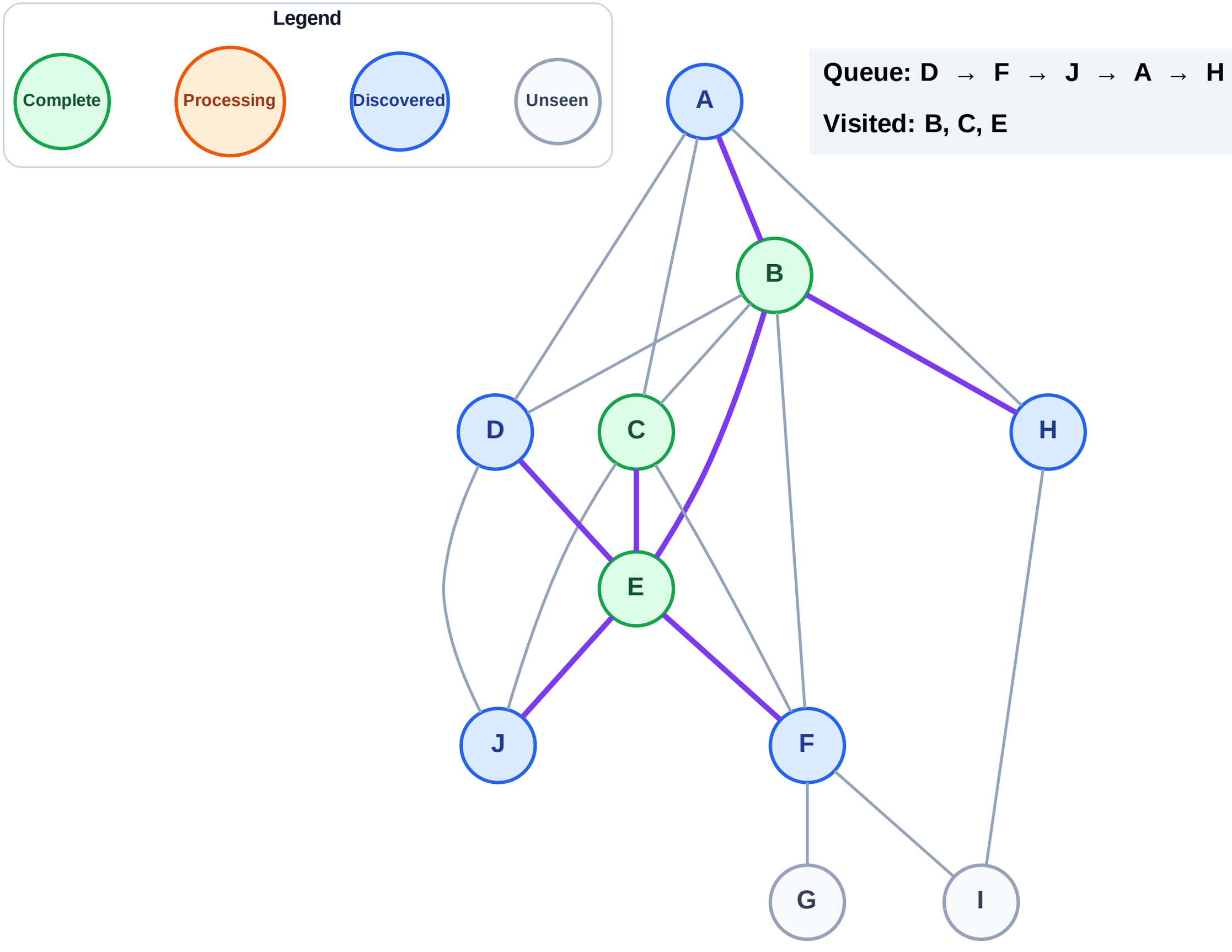


Queue: D → F → J → A → H

Visited: B, E

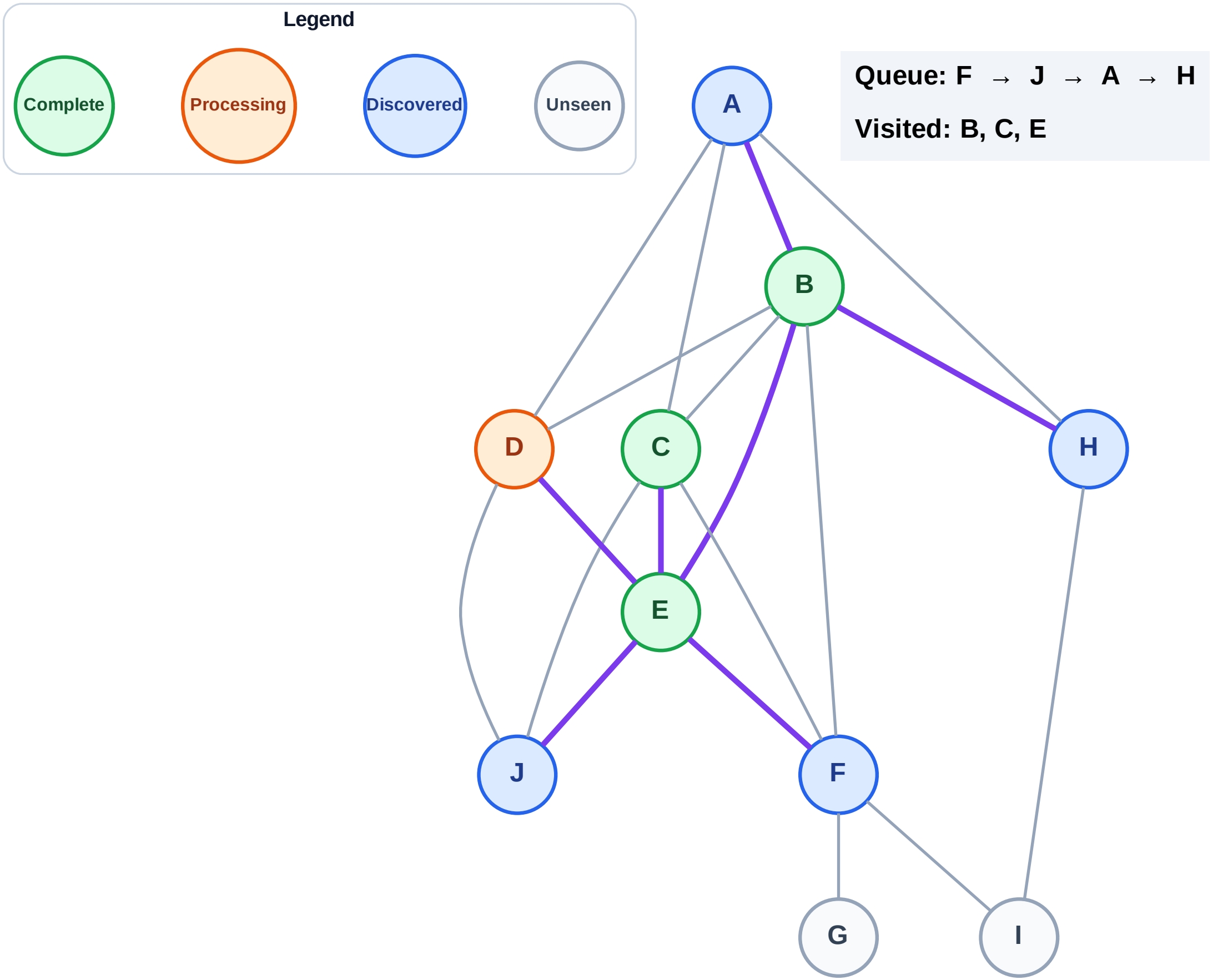
BFS Traversal

Step 7: No new neighbors from C



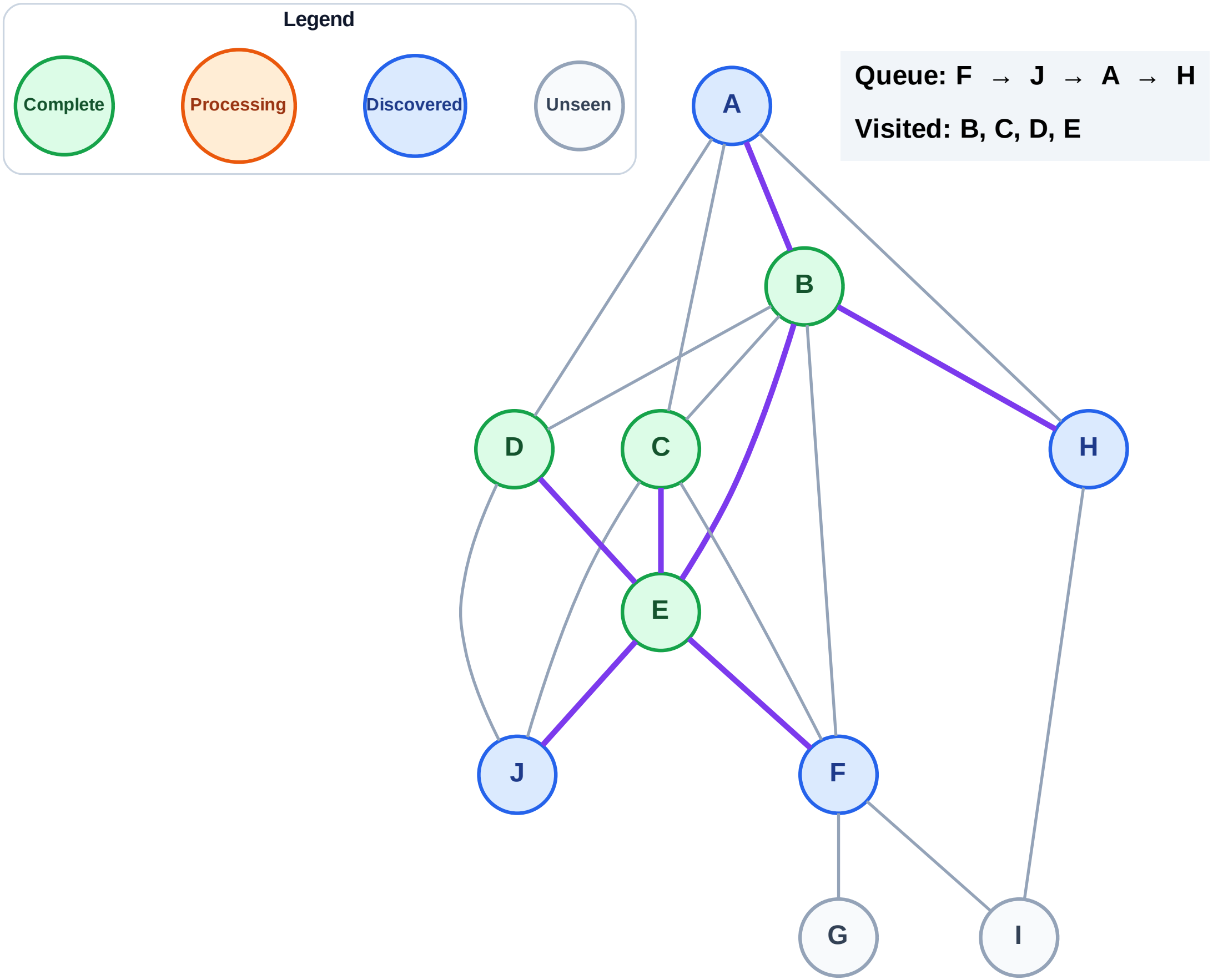
BFS Traversal

Step 8: Process node D



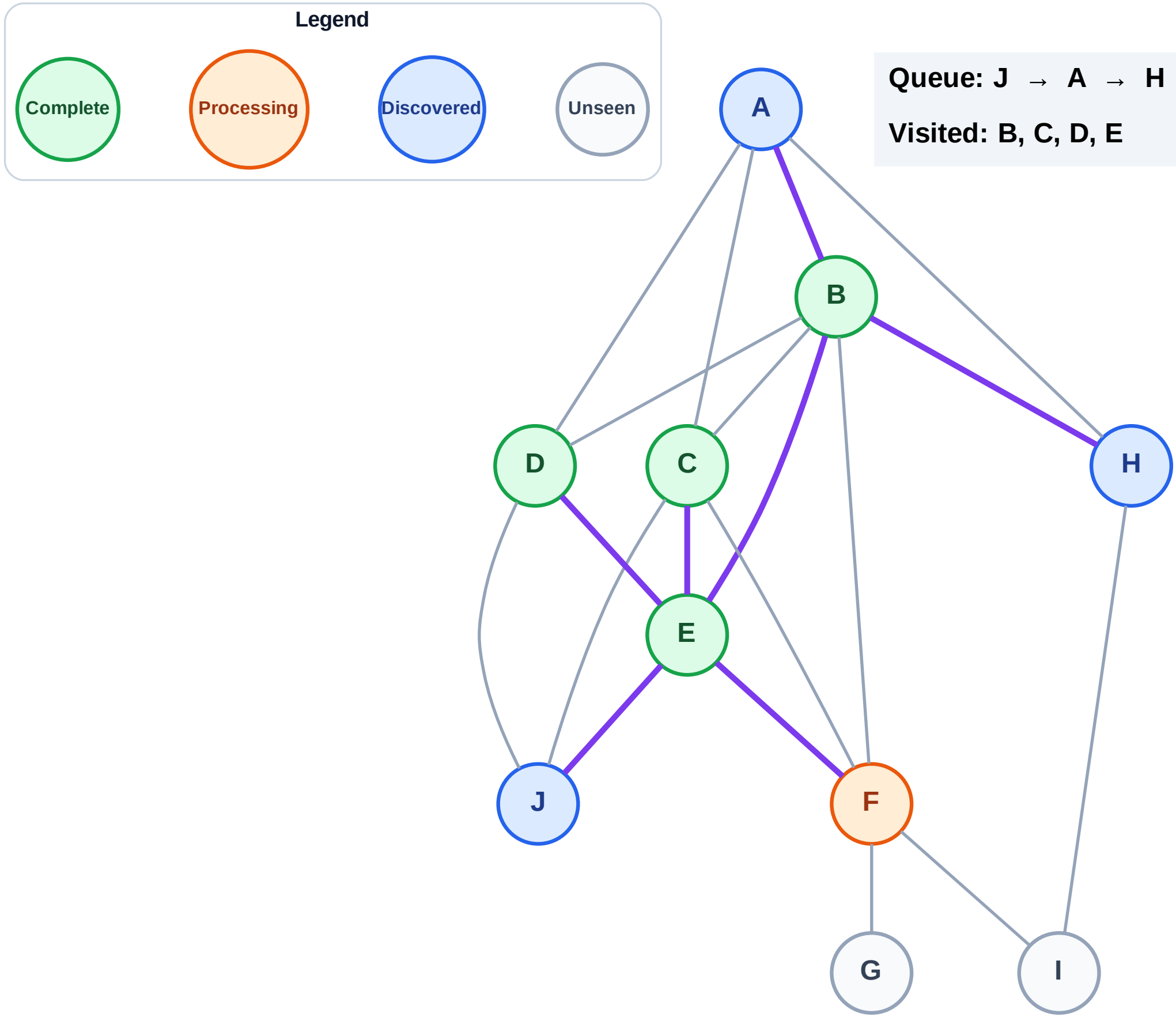
BFS Traversal

Step 9: No new neighbors from D



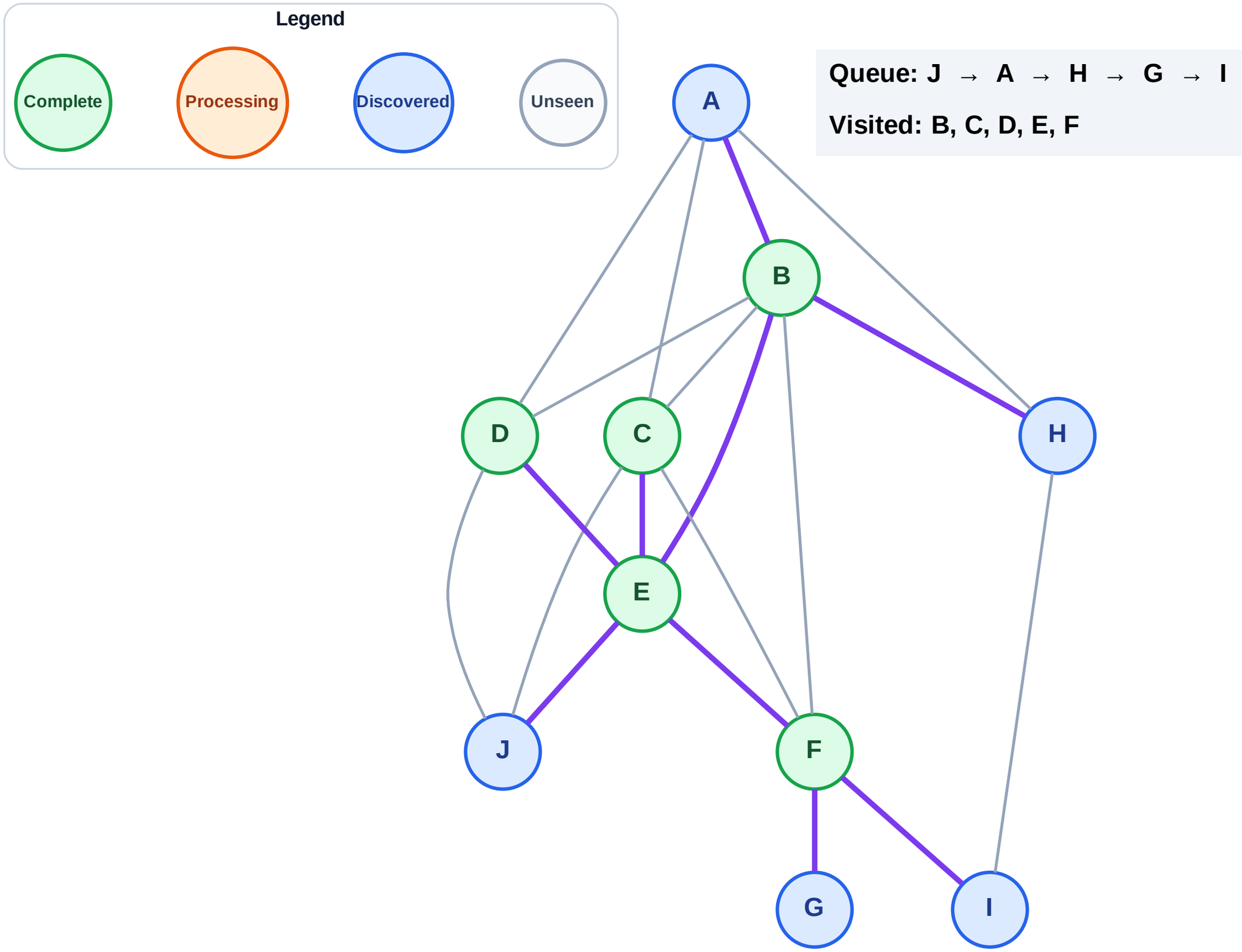
BFS Traversal

Step 10: Process node F



BFS Traversal

Step 11: Discover G, I from F

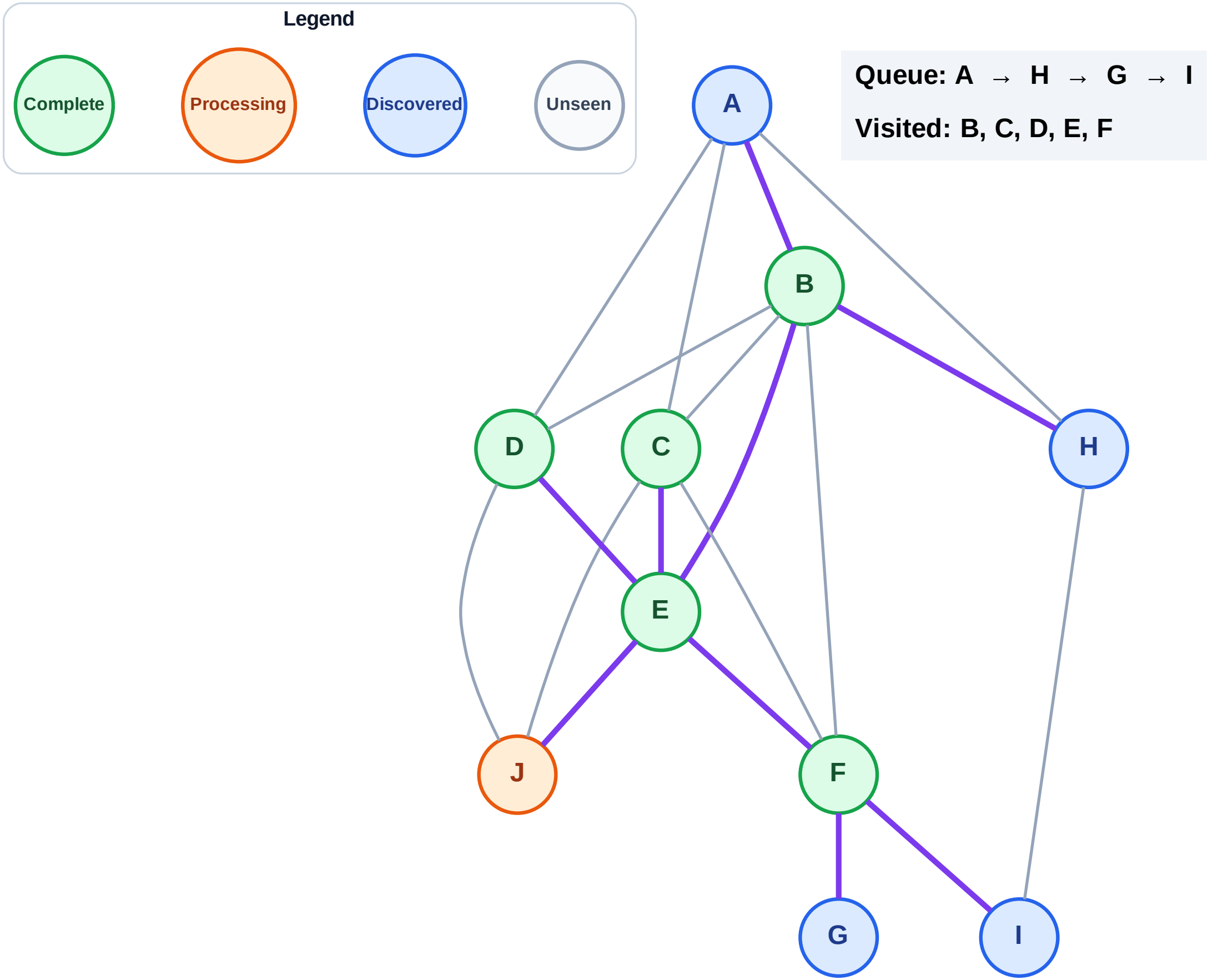


Queue: J → A → H → G → I

Visited: B, C, D, E, F

BFS Traversal

Step 12: Process node J

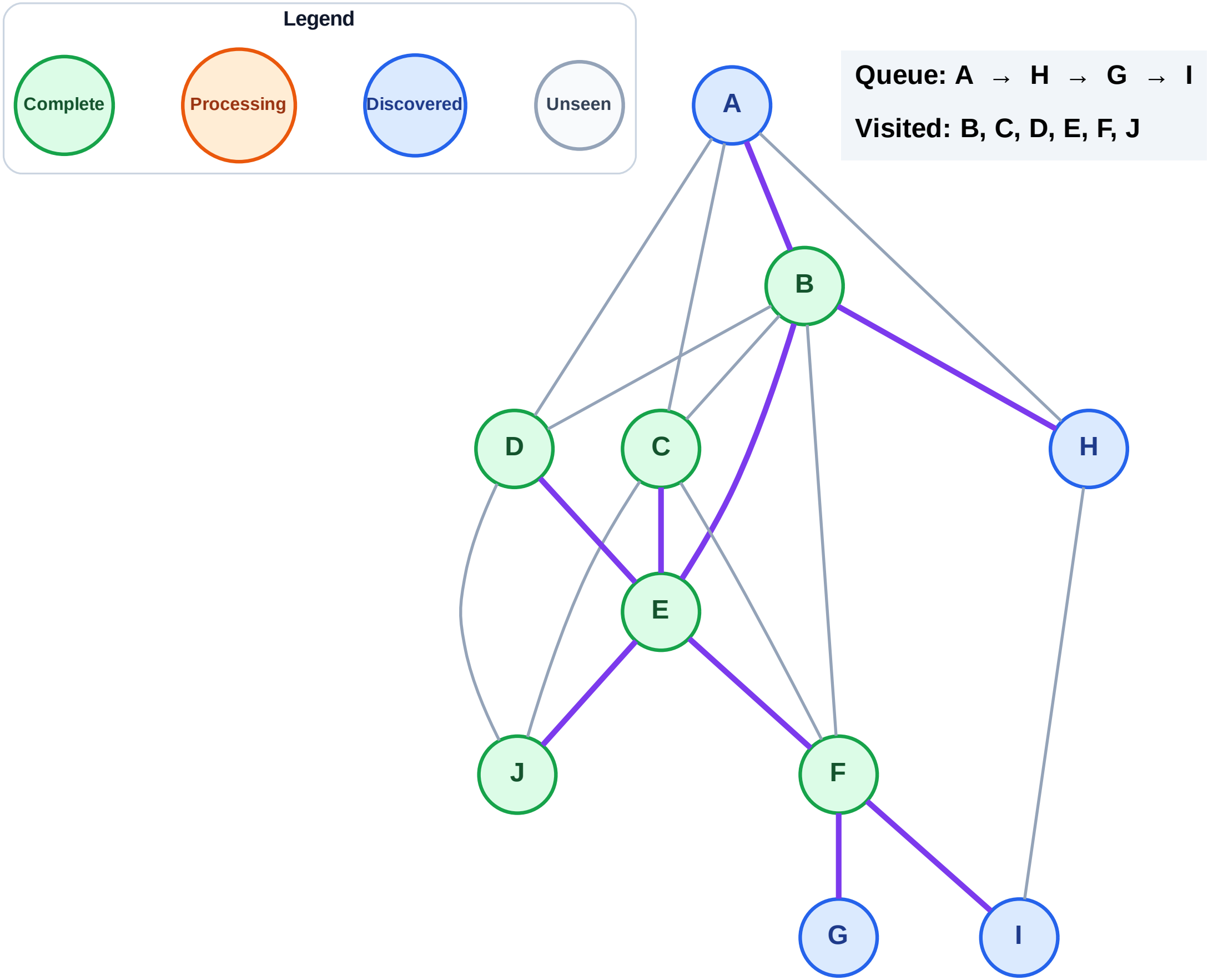


Queue: A → H → G → I

Visited: B, C, D, E, F

BFS Traversal

Step 13: No new neighbors from J

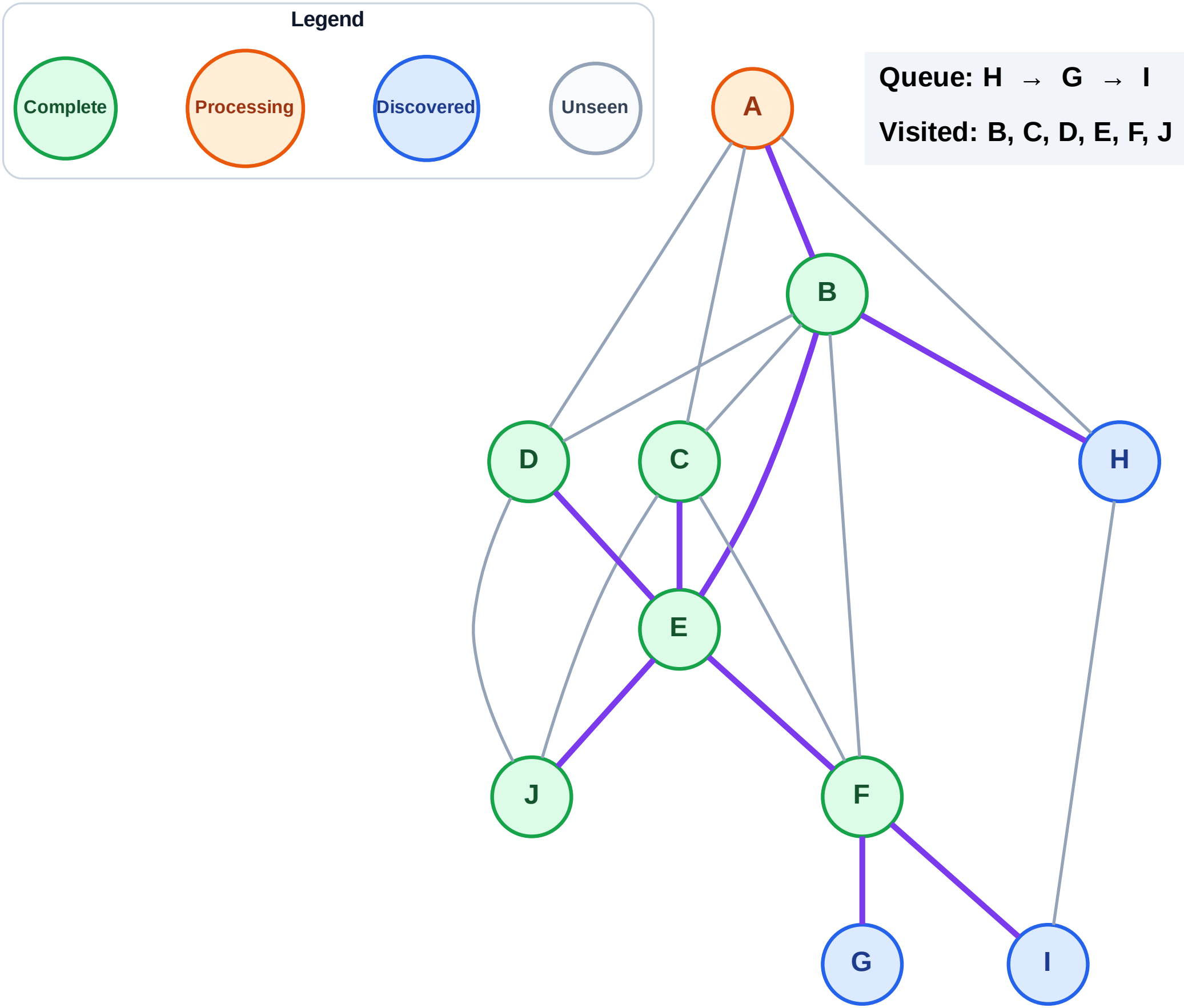


Queue: A → H → G → I

Visited: B, C, D, E, F, J

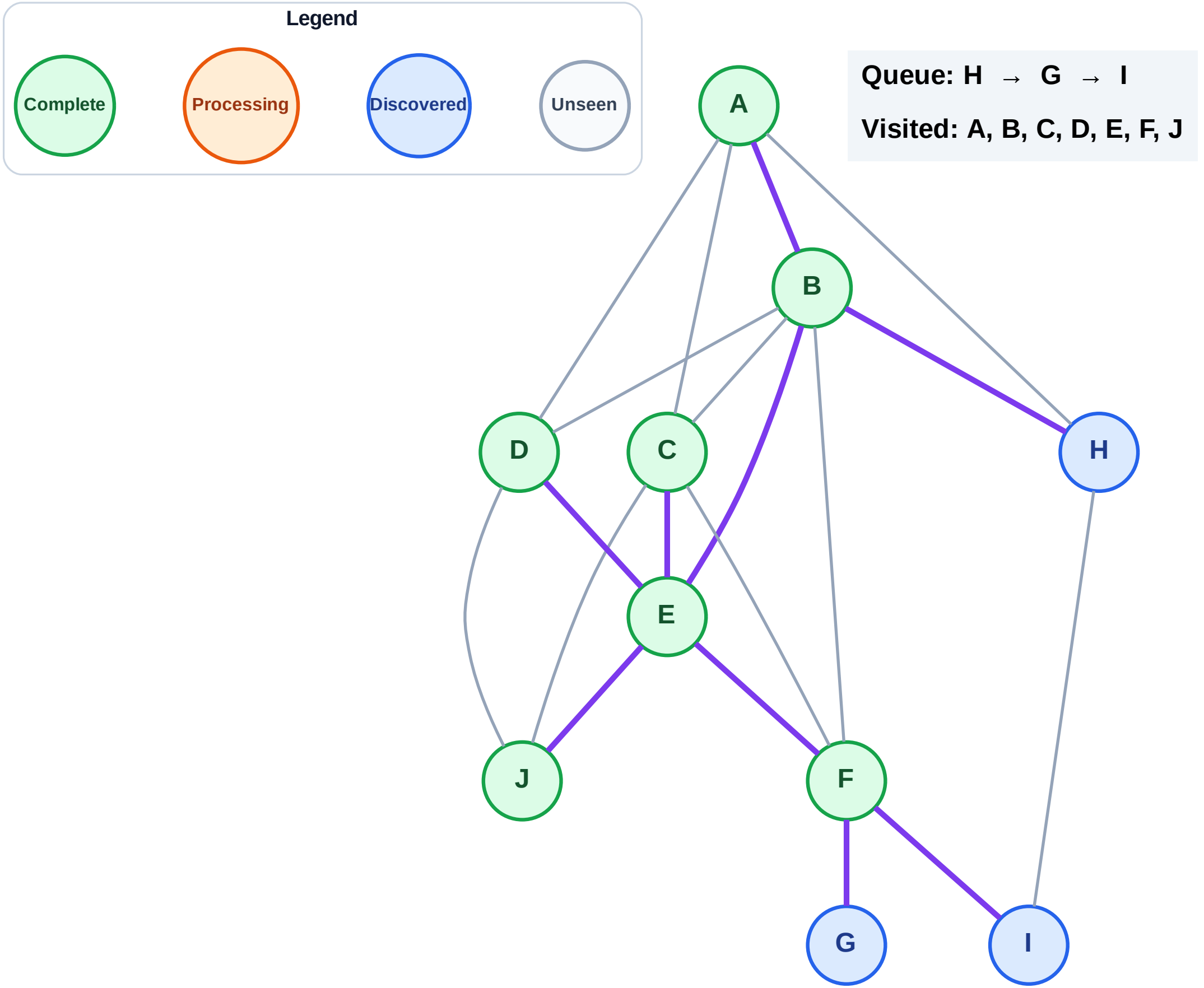
BFS Traversal

Step 14: Process node A



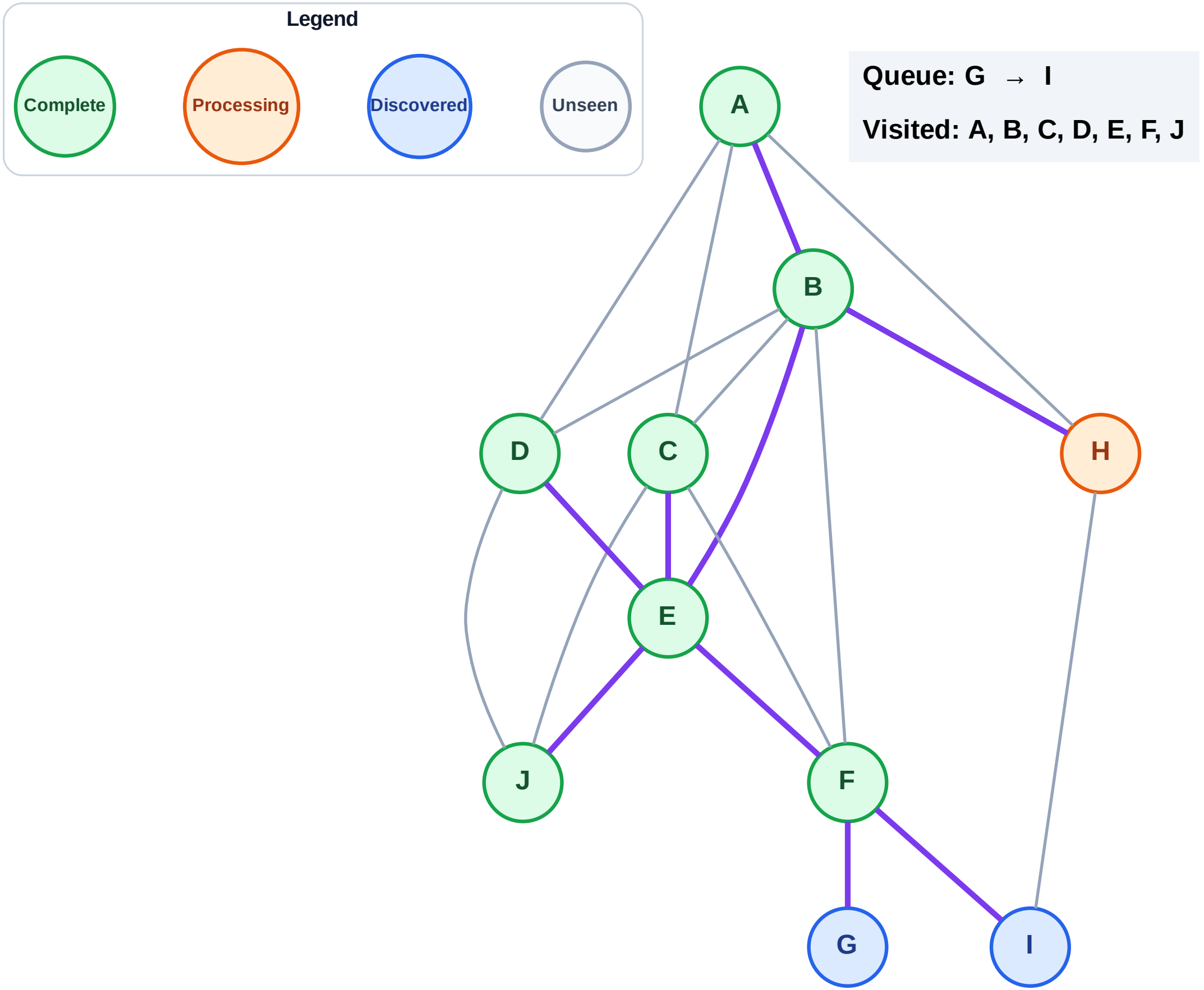
BFS Traversal

Step 15: No new neighbors from A



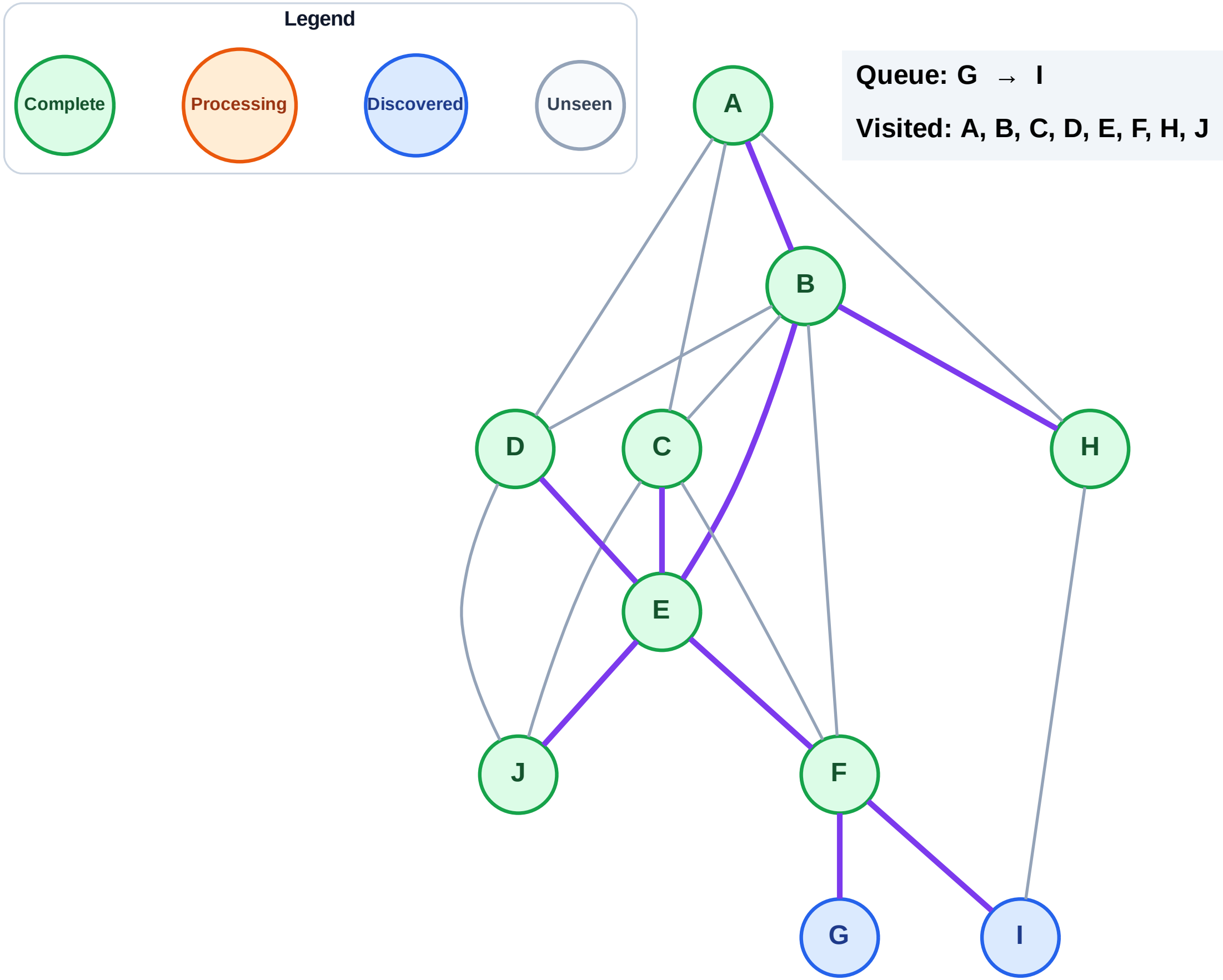
BFS Traversal

Step 16: Process node H



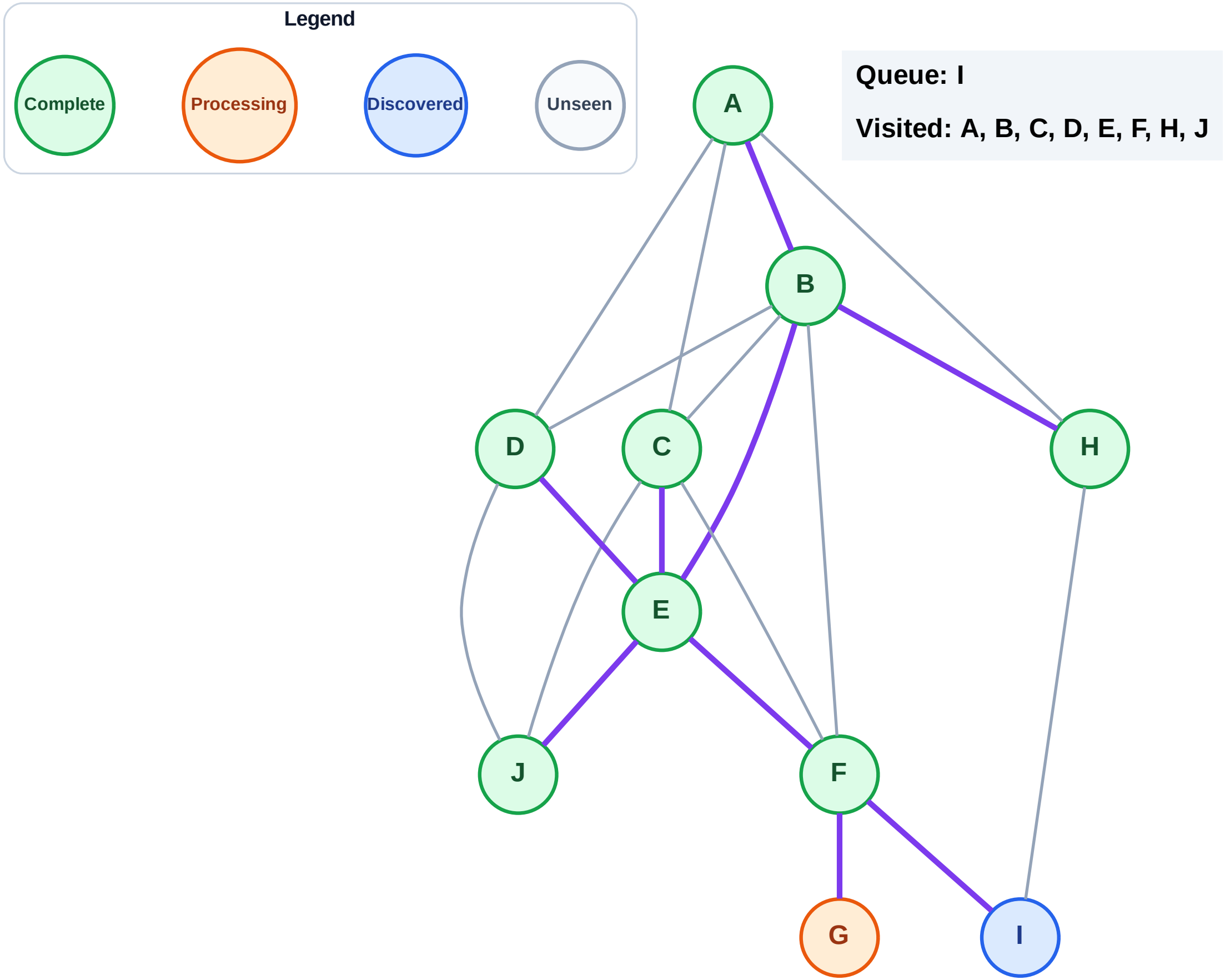
BFS Traversal

Step 17: No new neighbors from H



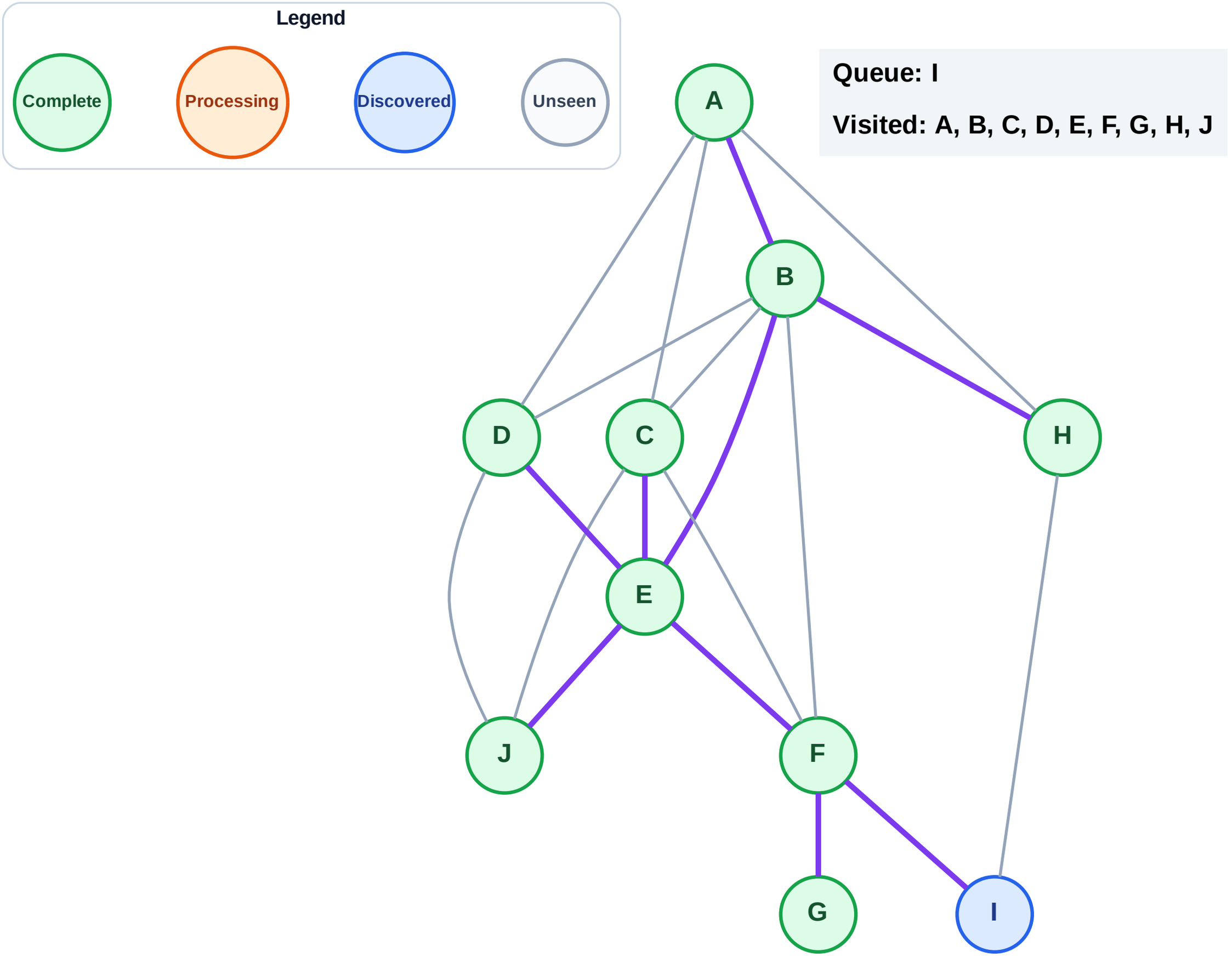
BFS Traversal

Step 18: Process node G



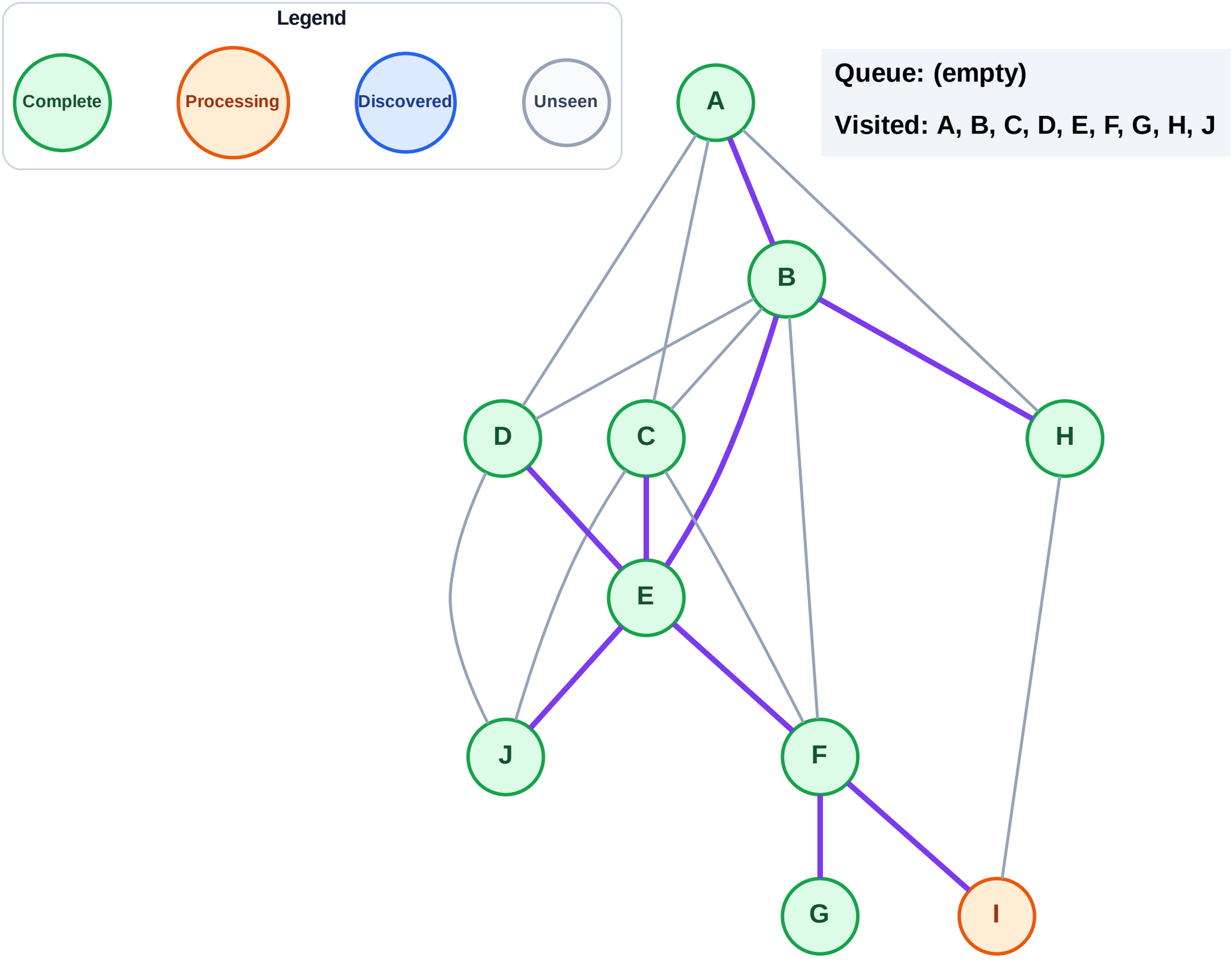
BFS Traversal

Step 19: No new neighbors from G



BFS Traversal

Step 20: Process node I

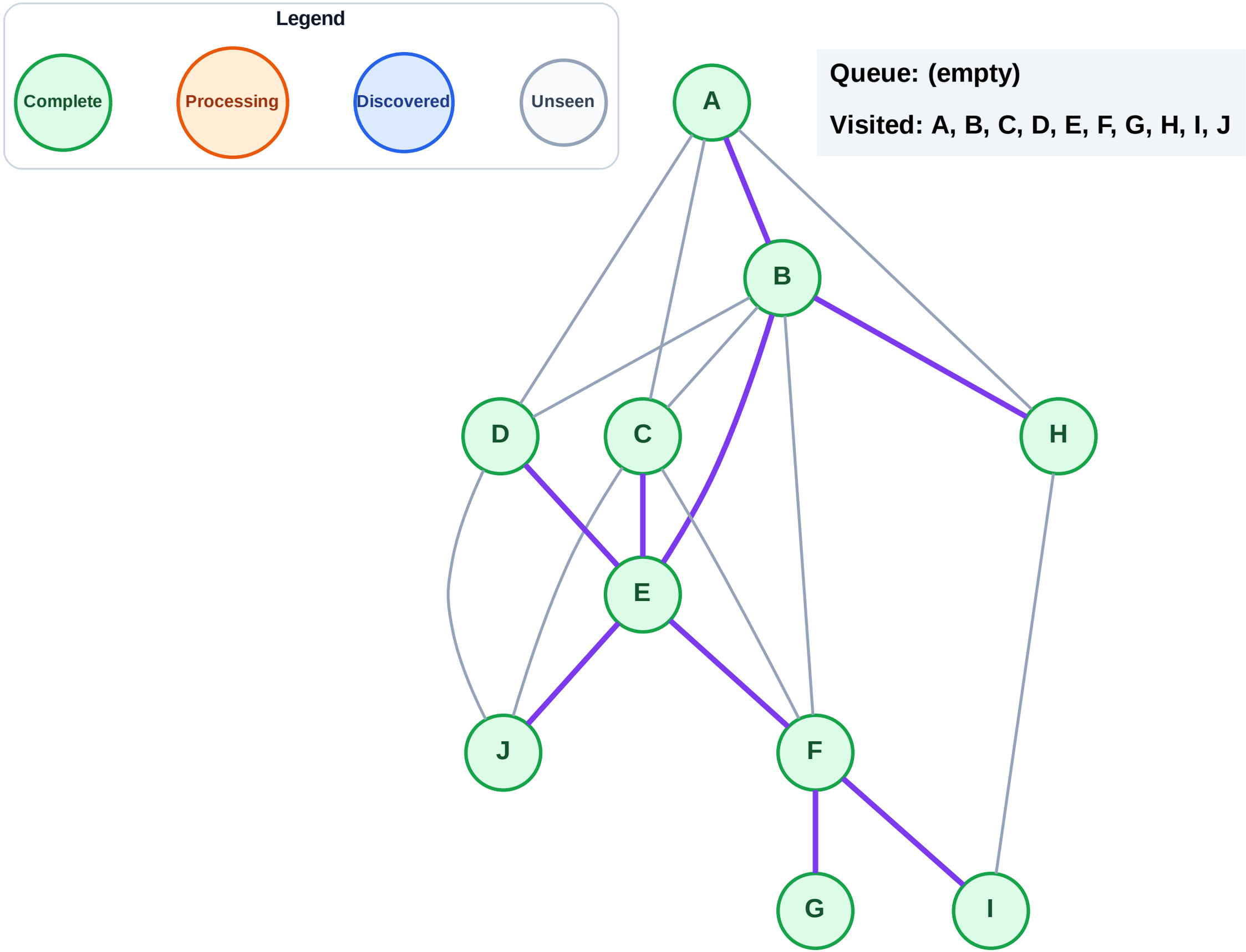


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J

BFS Traversal

Step 21: No new neighbors from I

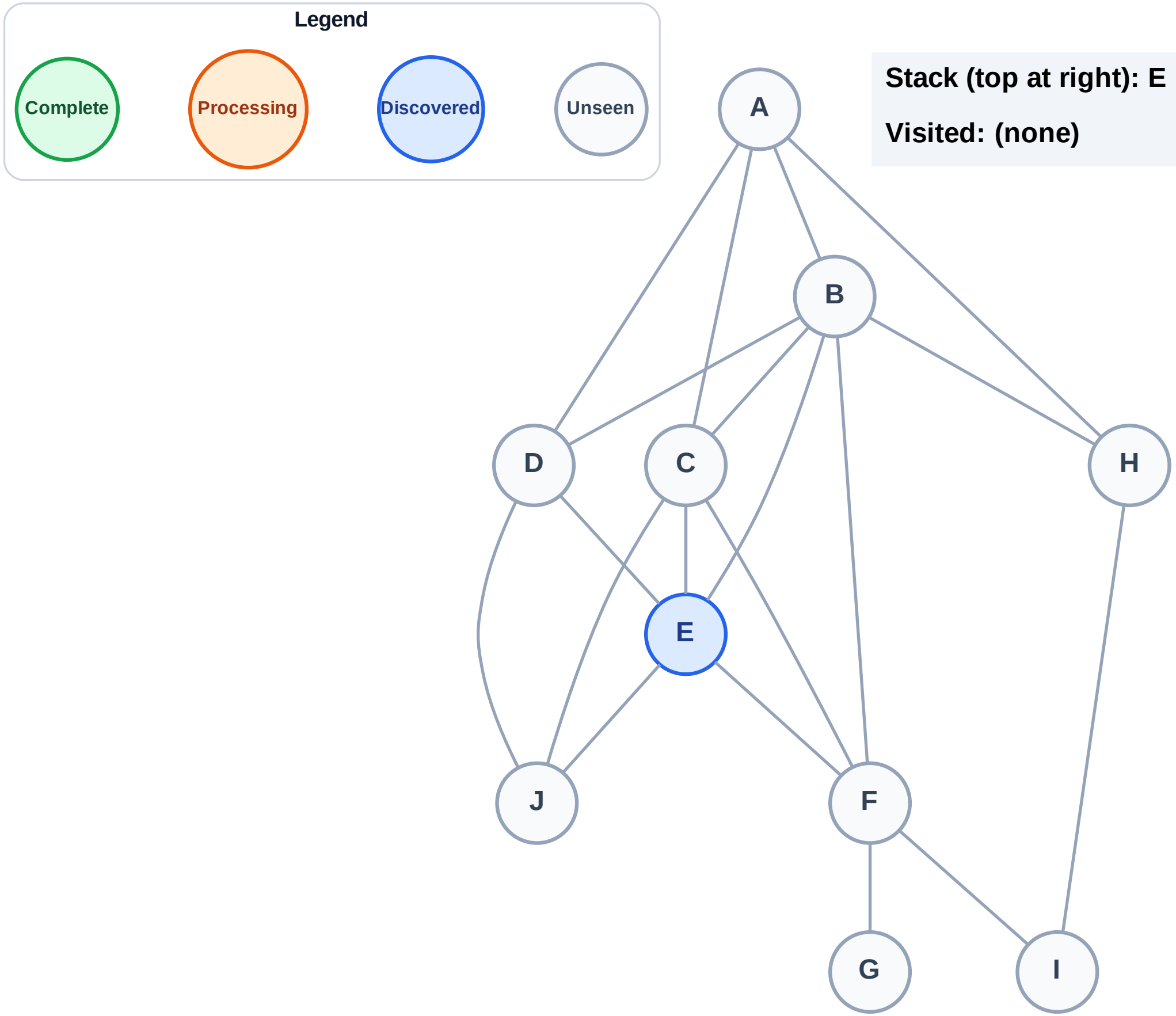


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

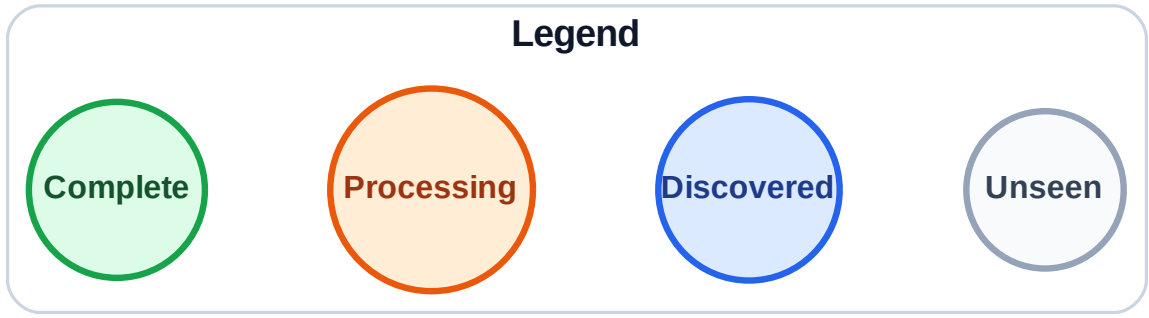
DFS Traversal

Step 1: Start at E



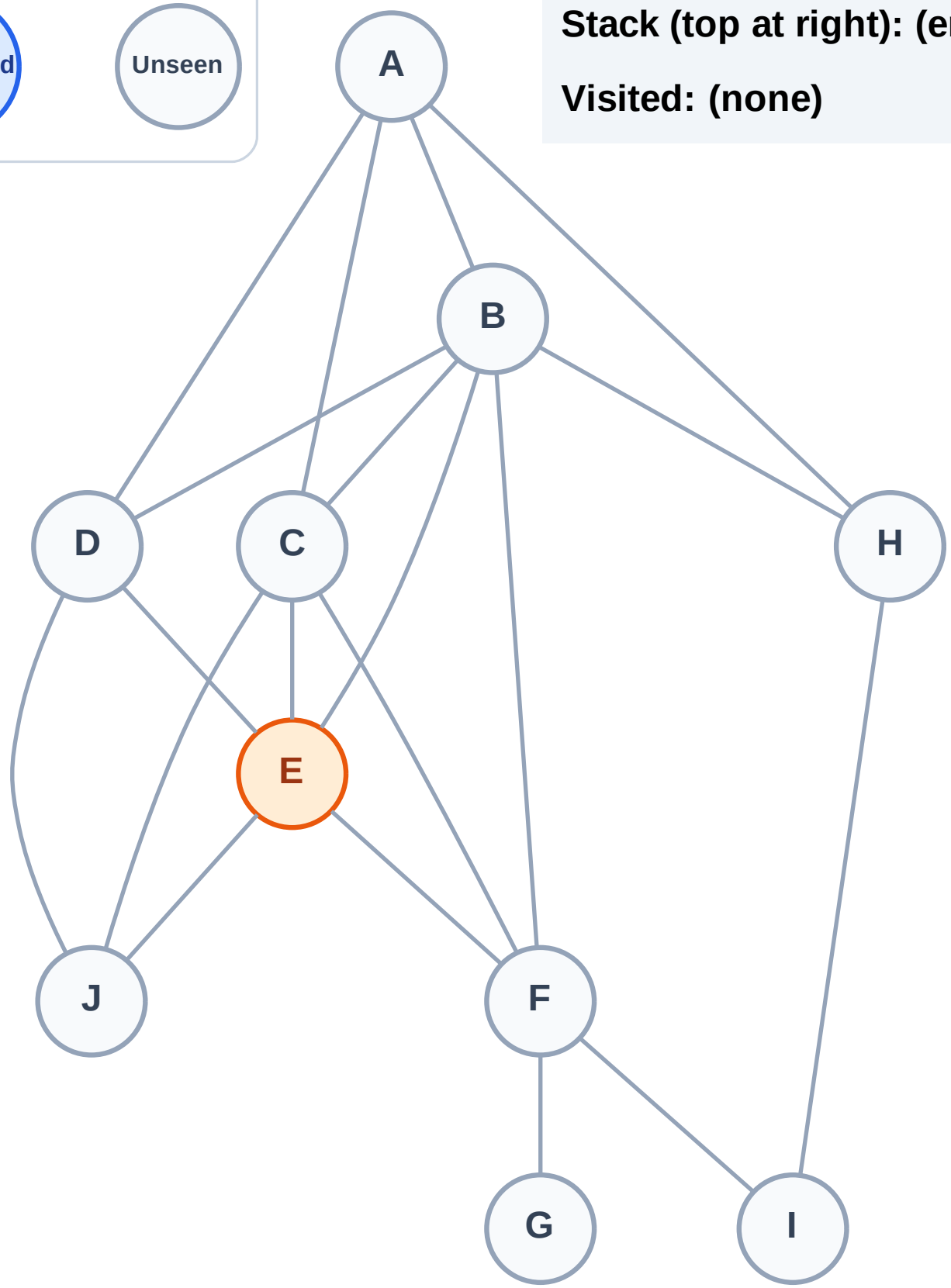
DFS Traversal

Step 2: Process node E



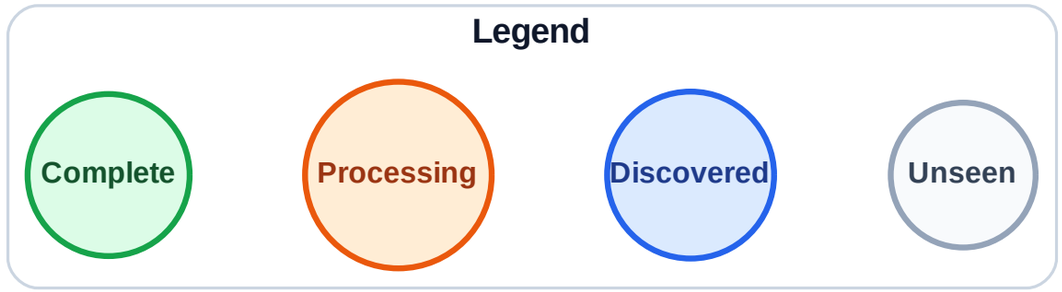
Stack (top at right): (empty)

Visited: (none)



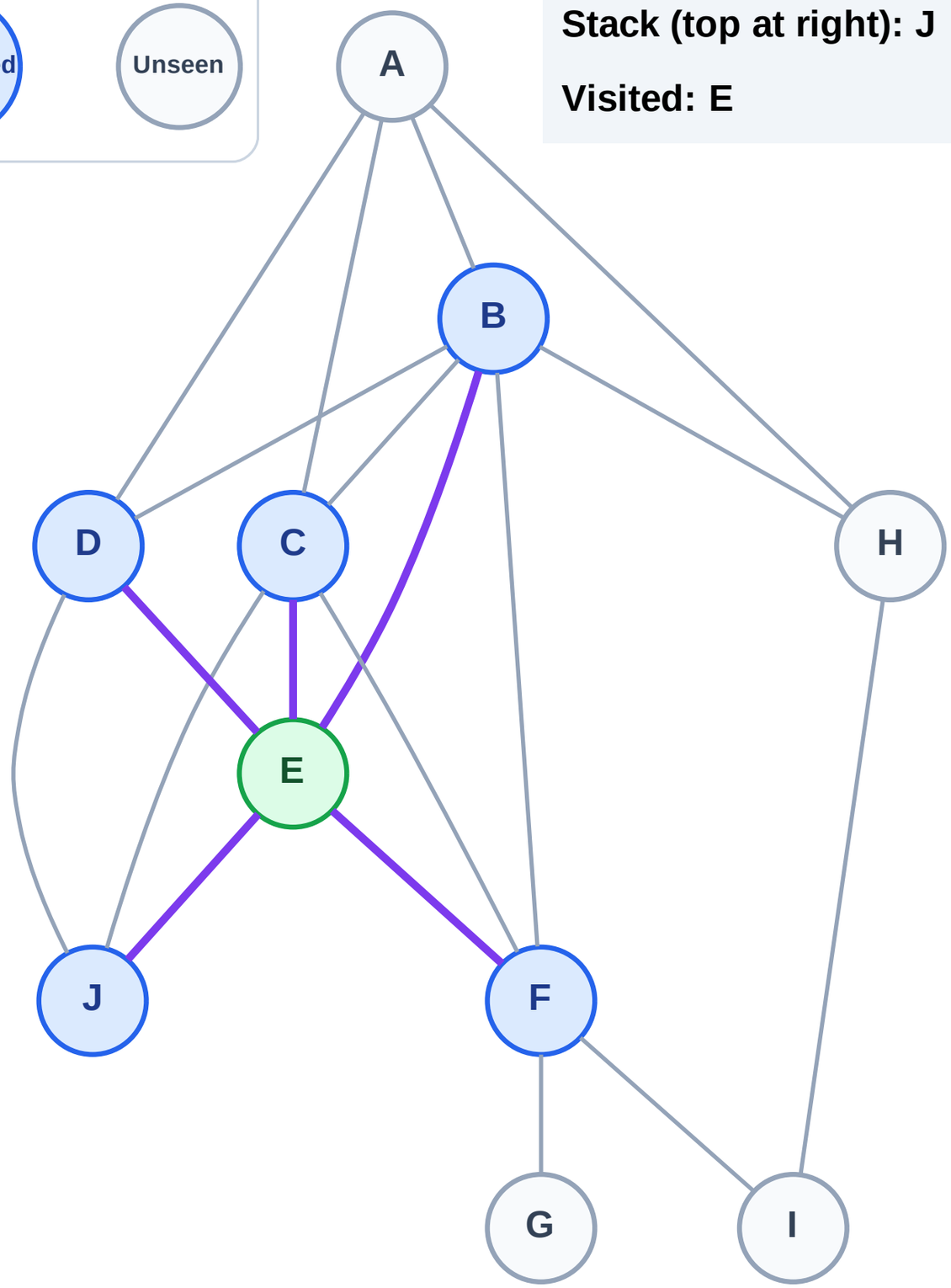
DFS Traversal

Step 3: Discover J, F, D, C, B from E



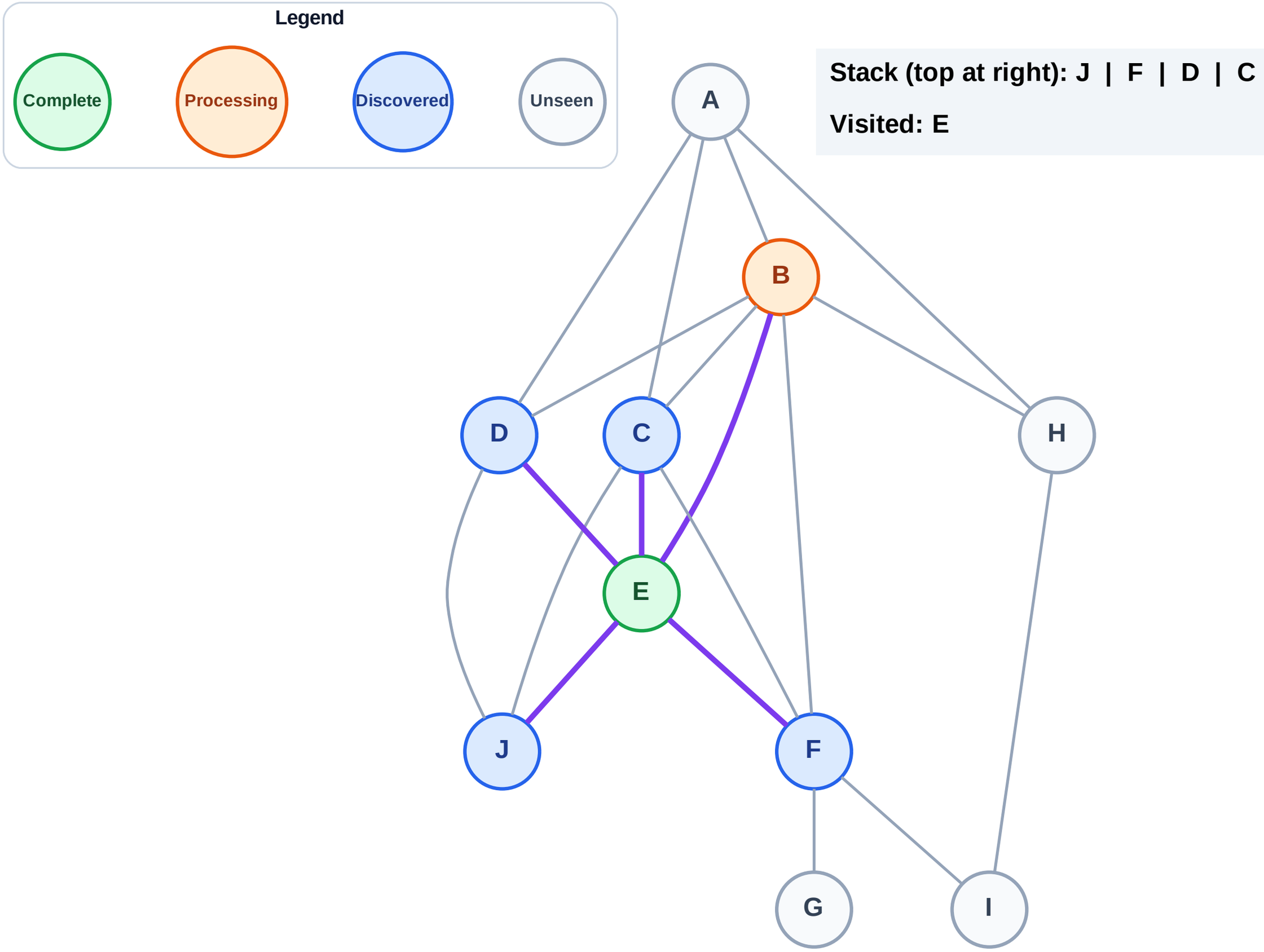
Stack (top at right): J | F | D | C | B

Visited: E



DFS Traversal

Step 4: Process node B

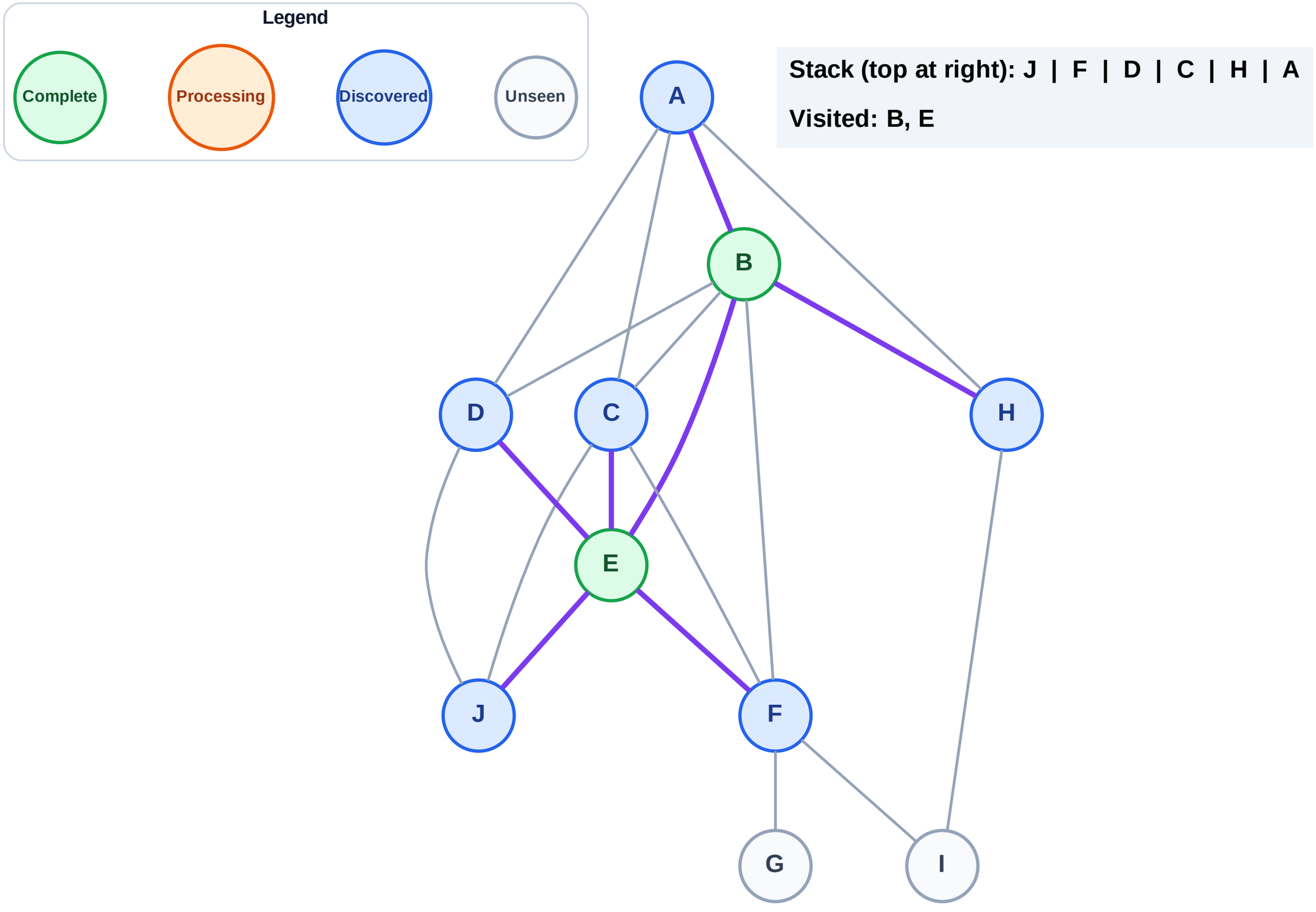


Stack (top at right): J | F | D | C

Visited: E

DFS Traversal

Step 5: Discover H, A from B

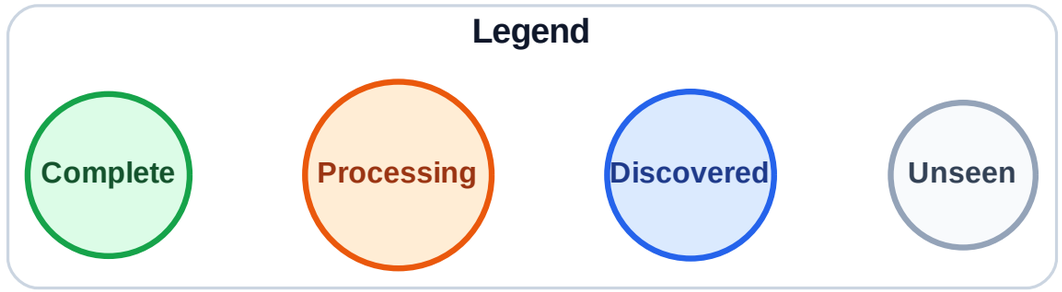


Stack (top at right): J | F | D | C | H | A

Visited: B, E

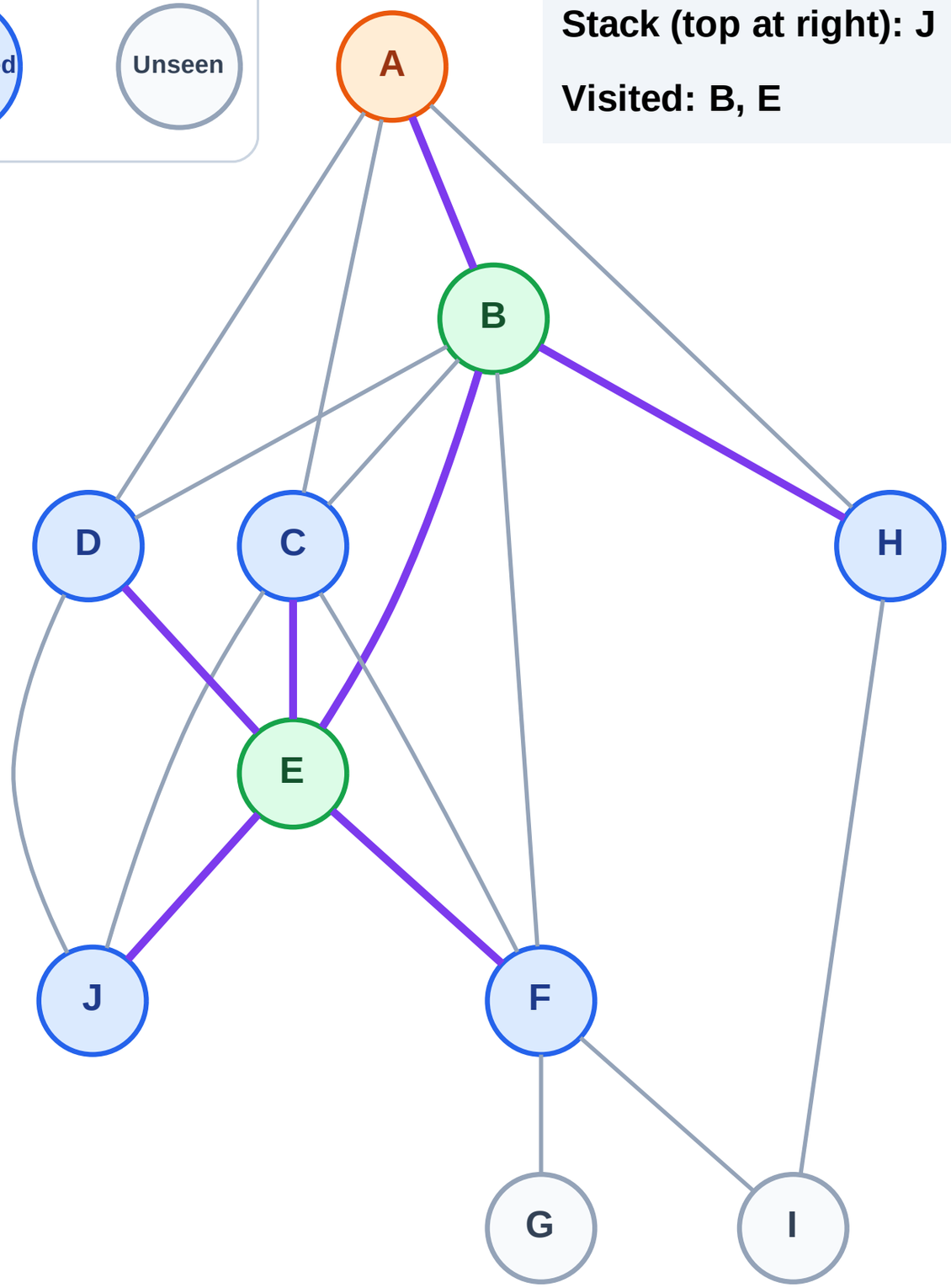
DFS Traversal

Step 6: Process node A



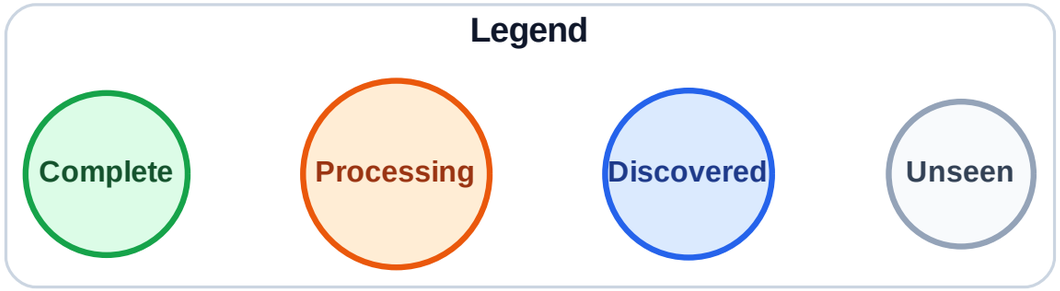
Stack (top at right): J | F | D | C | H

Visited: B, E



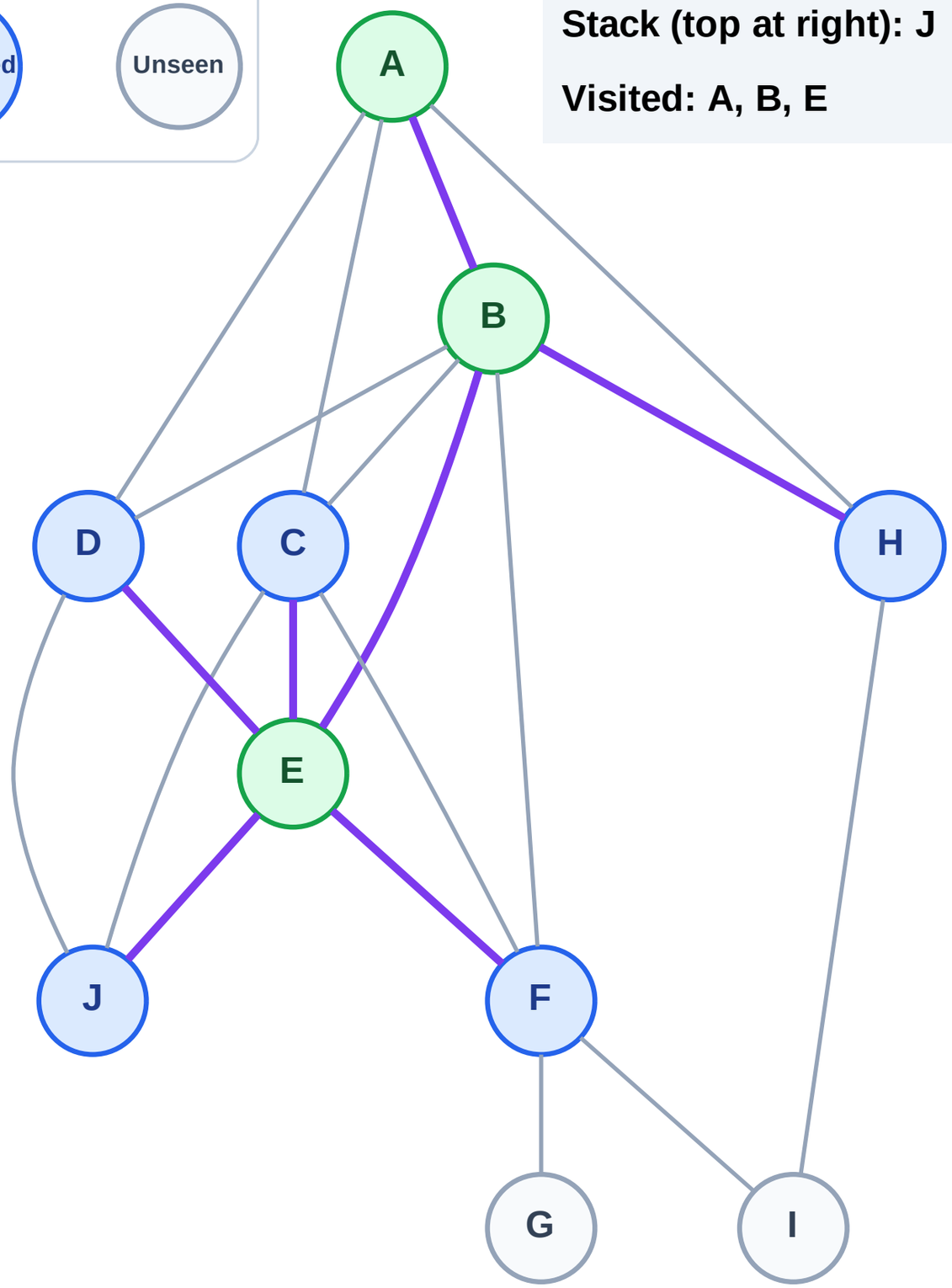
DFS Traversal

Step 7: No new neighbors from A



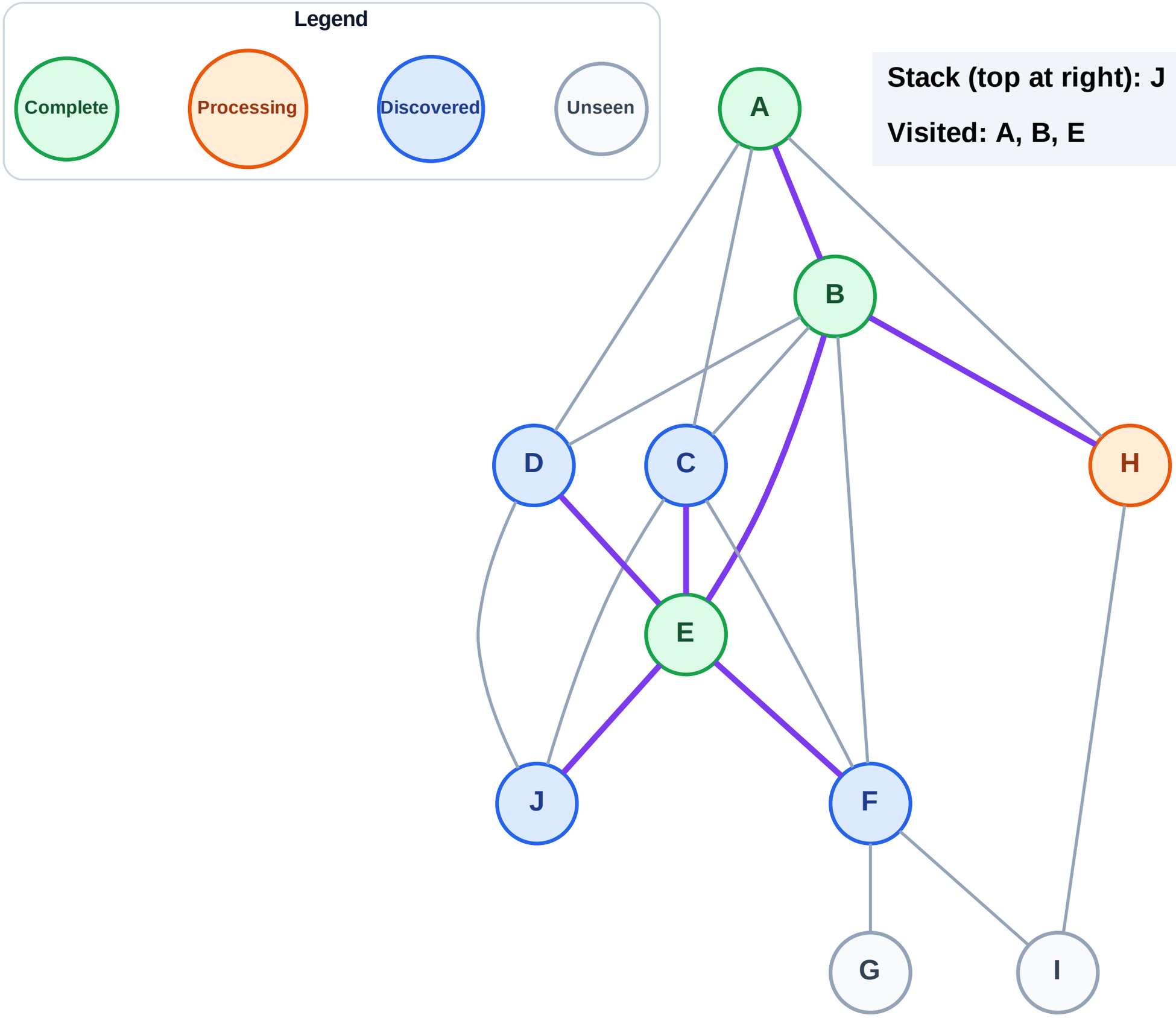
Stack (top at right): J | F | D | C | H

Visited: A, B, E



DFS Traversal

Step 8: Process node H

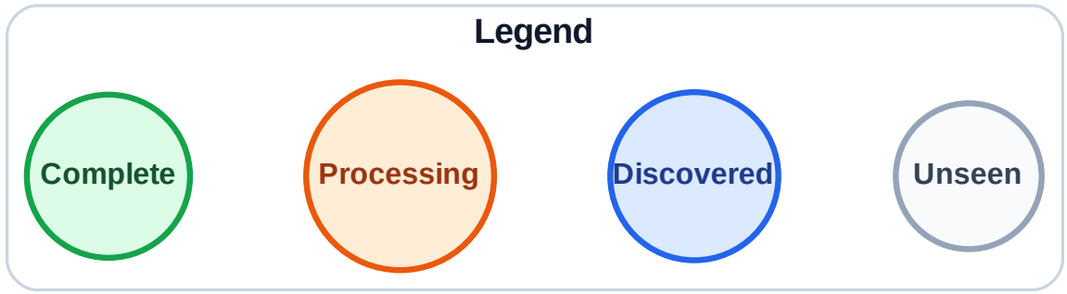


Stack (top at right): J | F | D | C

Visited: A, B, E

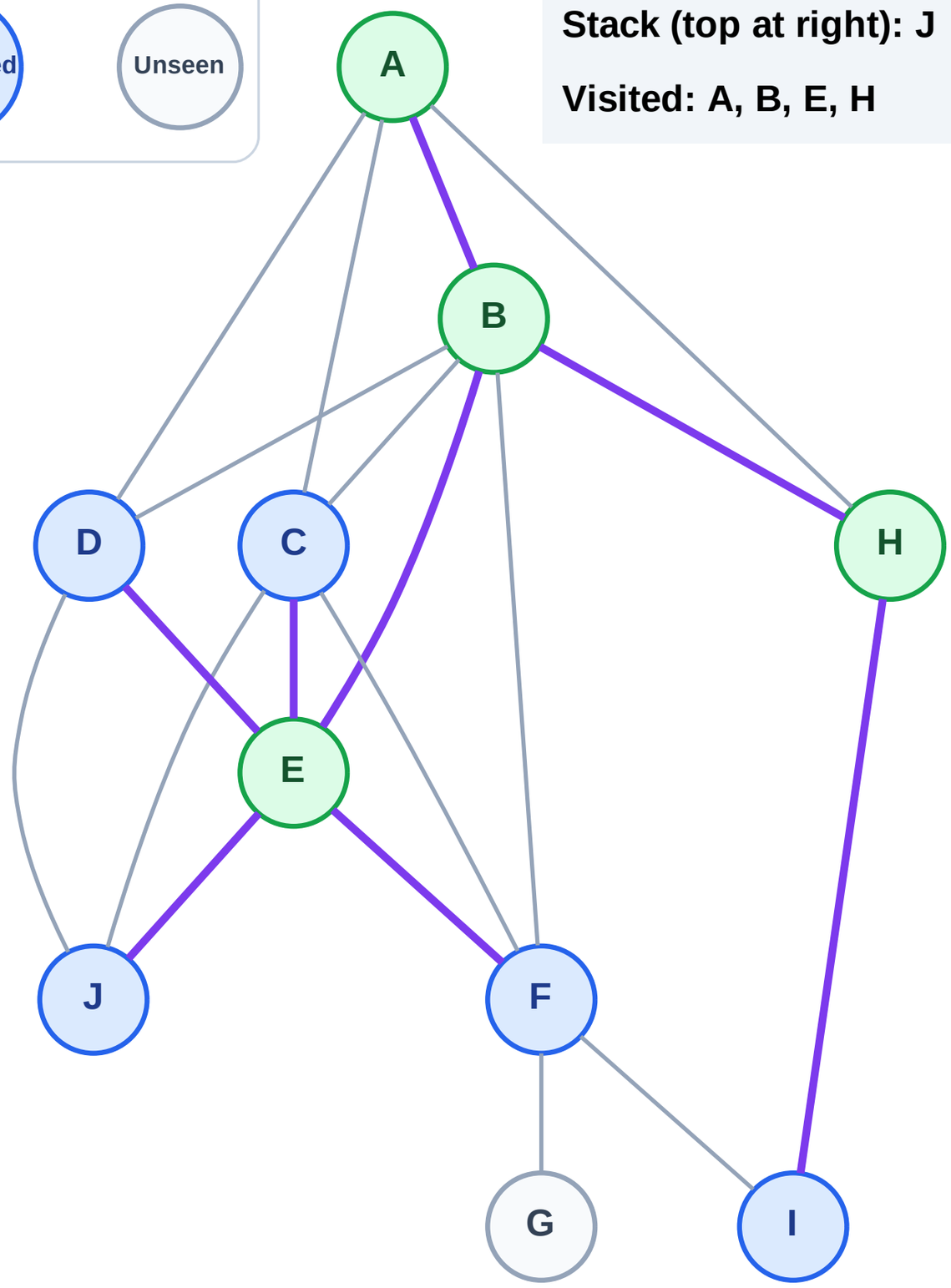
DFS Traversal

Step 9: Discover I from H



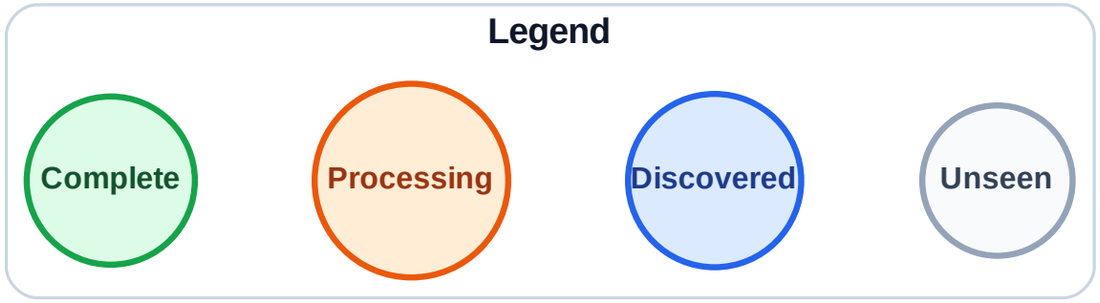
Stack (top at right): J | F | D | C | I

Visited: A, B, E, H



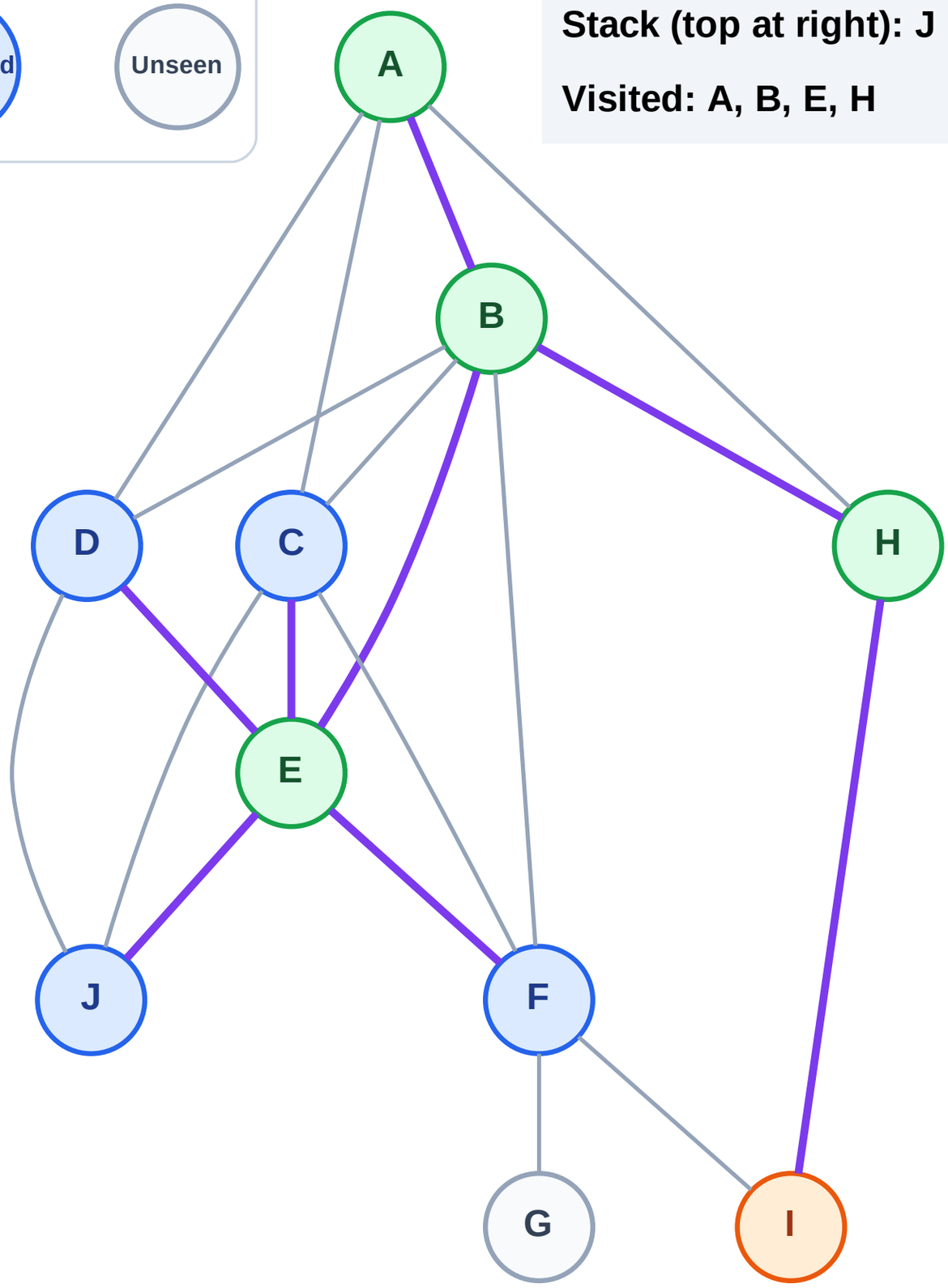
DFS Traversal

Step 10: Process node I



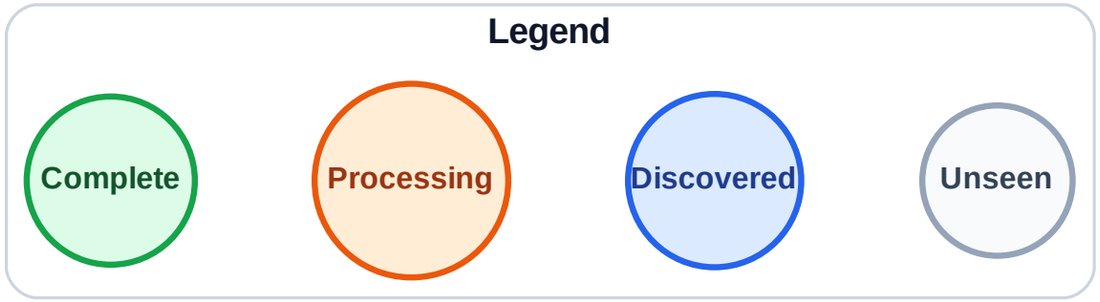
Stack (top at right): J | F | D | C

Visited: A, B, E, H



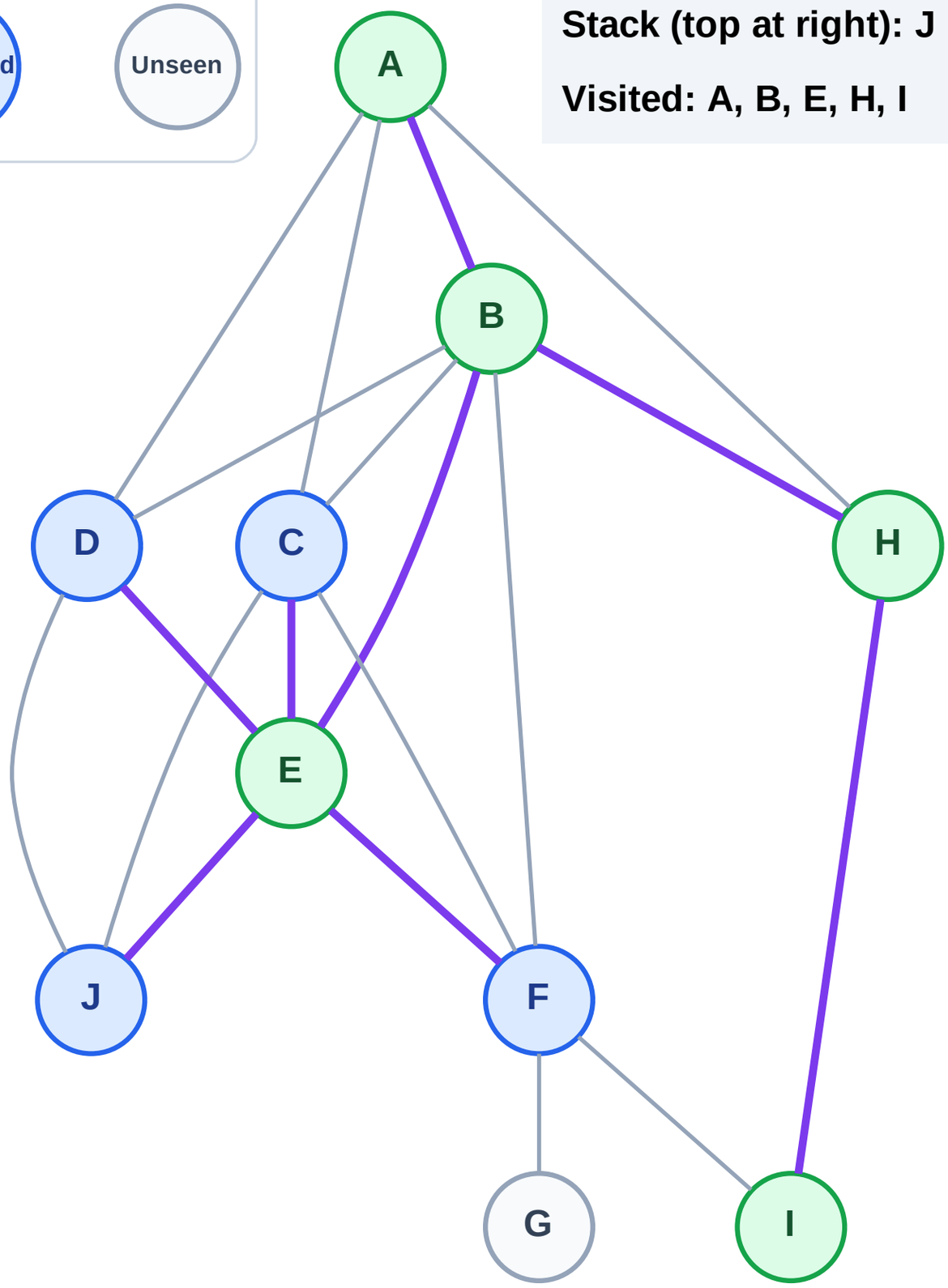
DFS Traversal

Step 11: No new neighbors from I



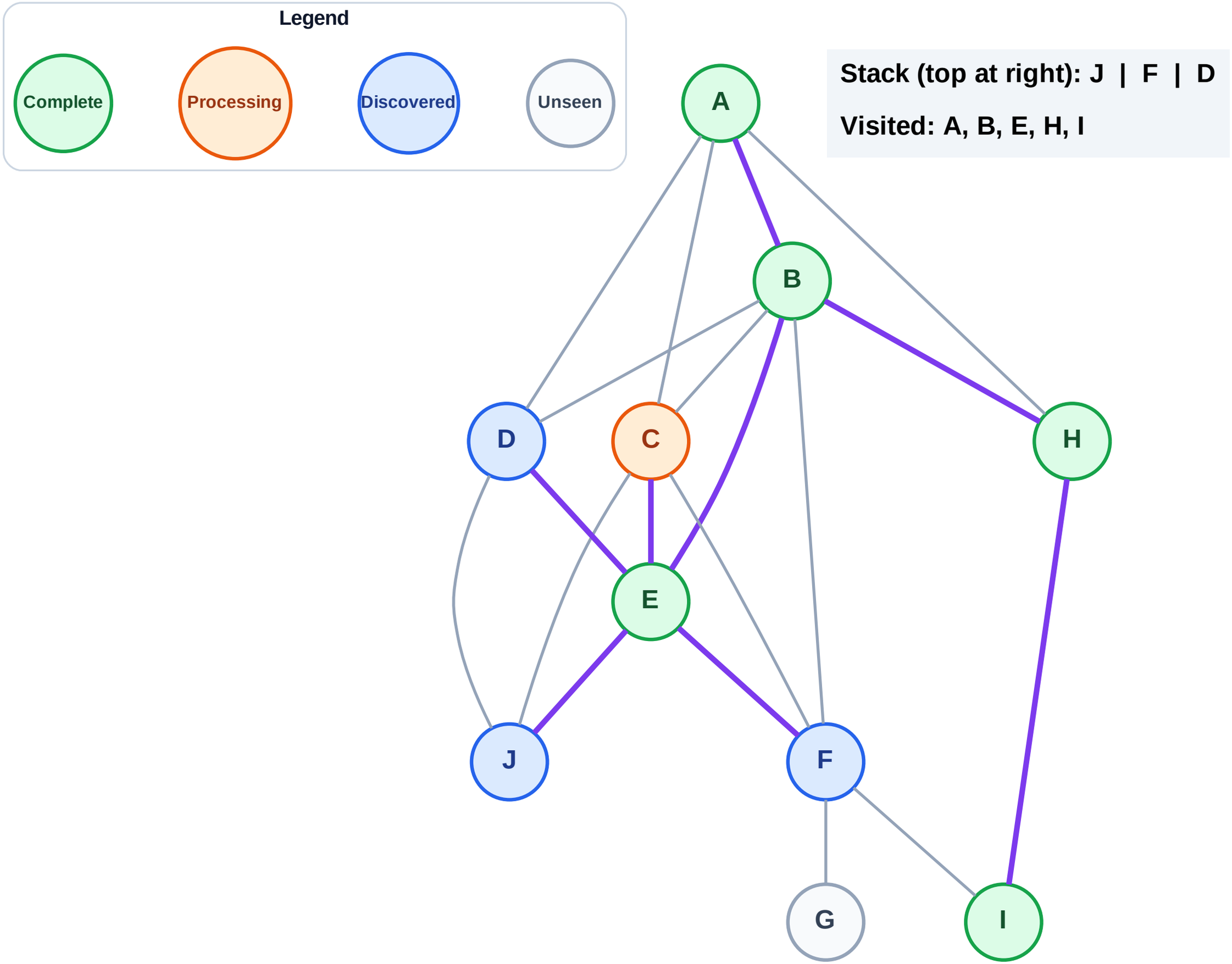
Stack (top at right): J | F | D | C

Visited: A, B, E, H, I



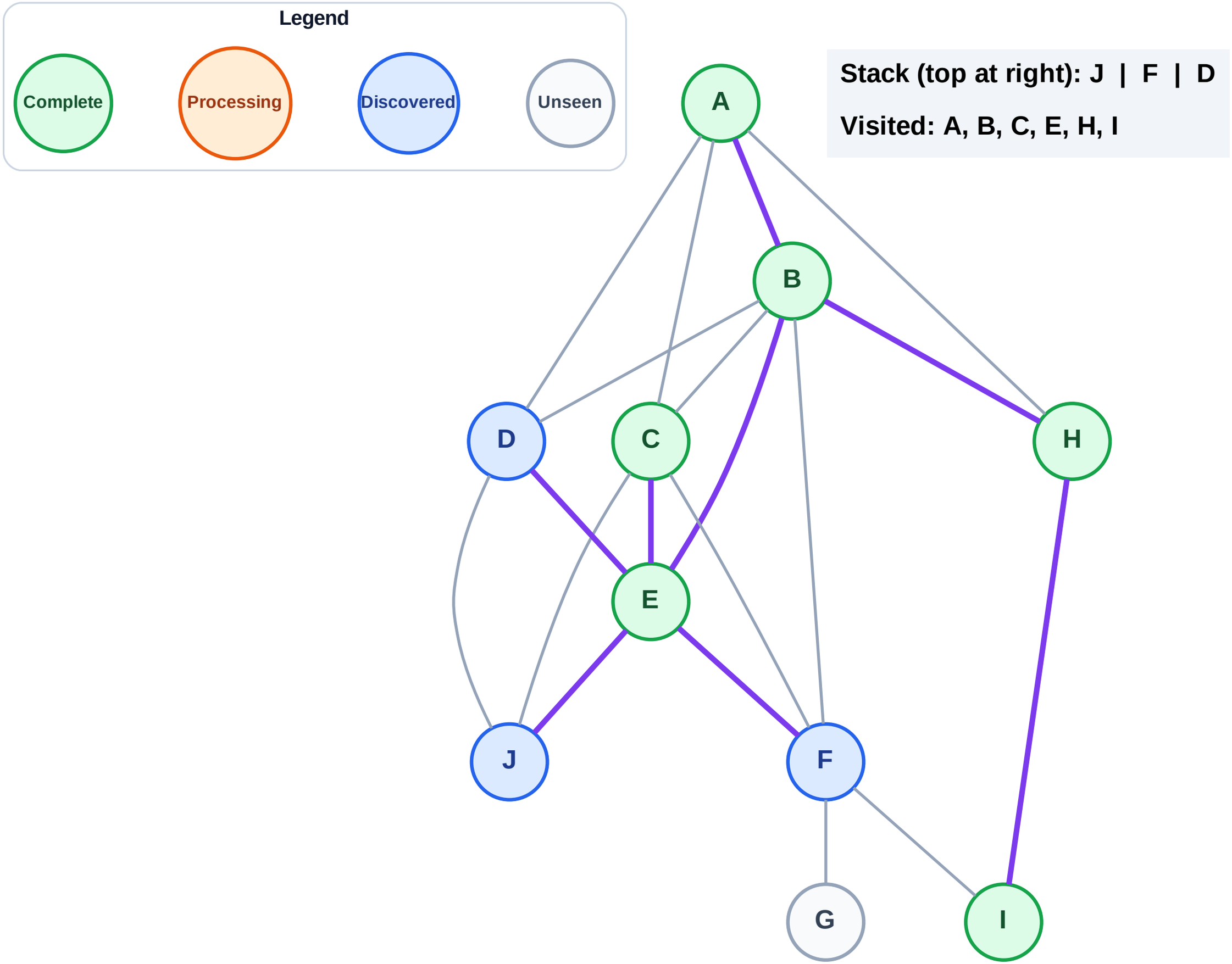
DFS Traversal

Step 12: Process node C



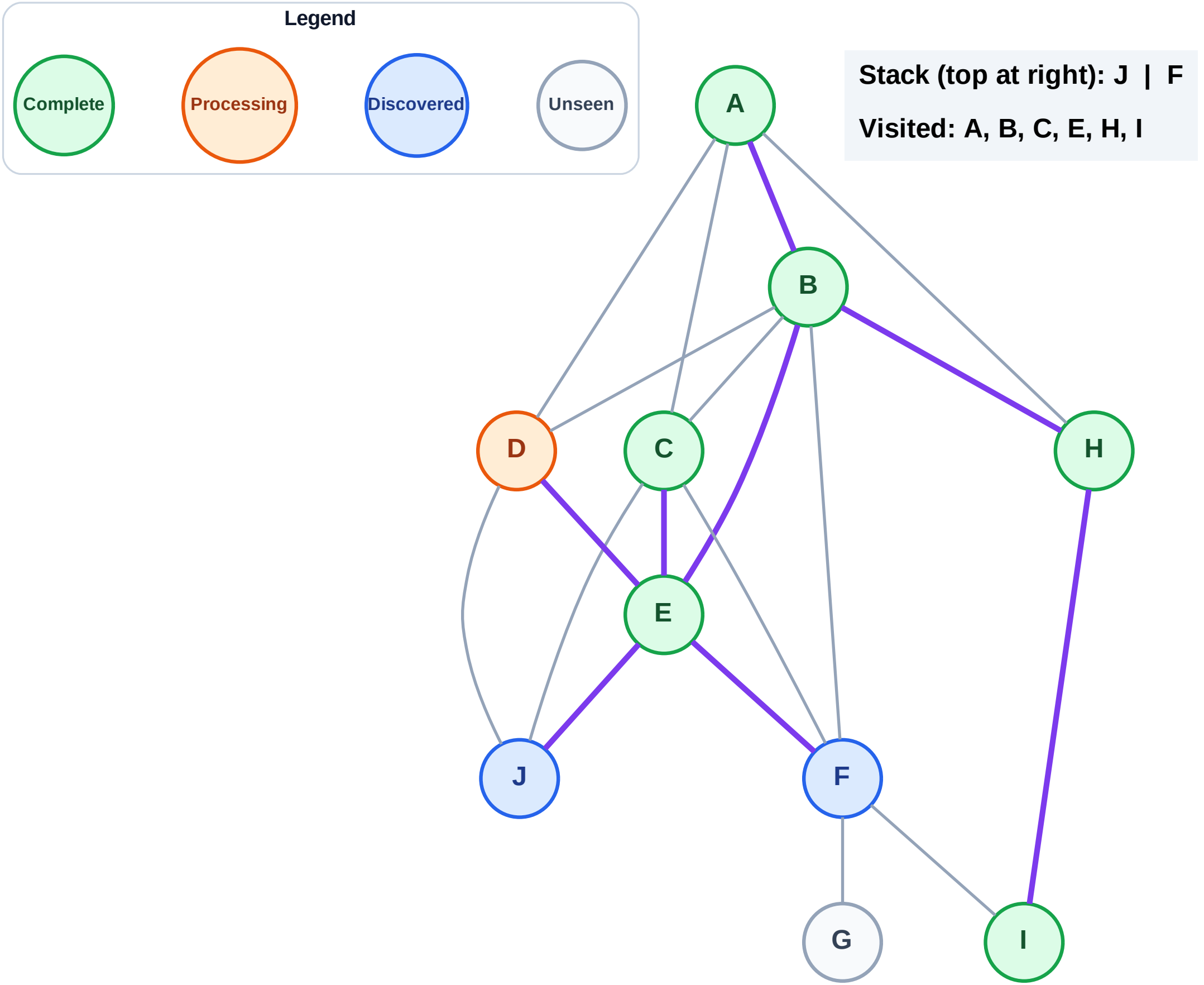
DFS Traversal

Step 13: No new neighbors from C



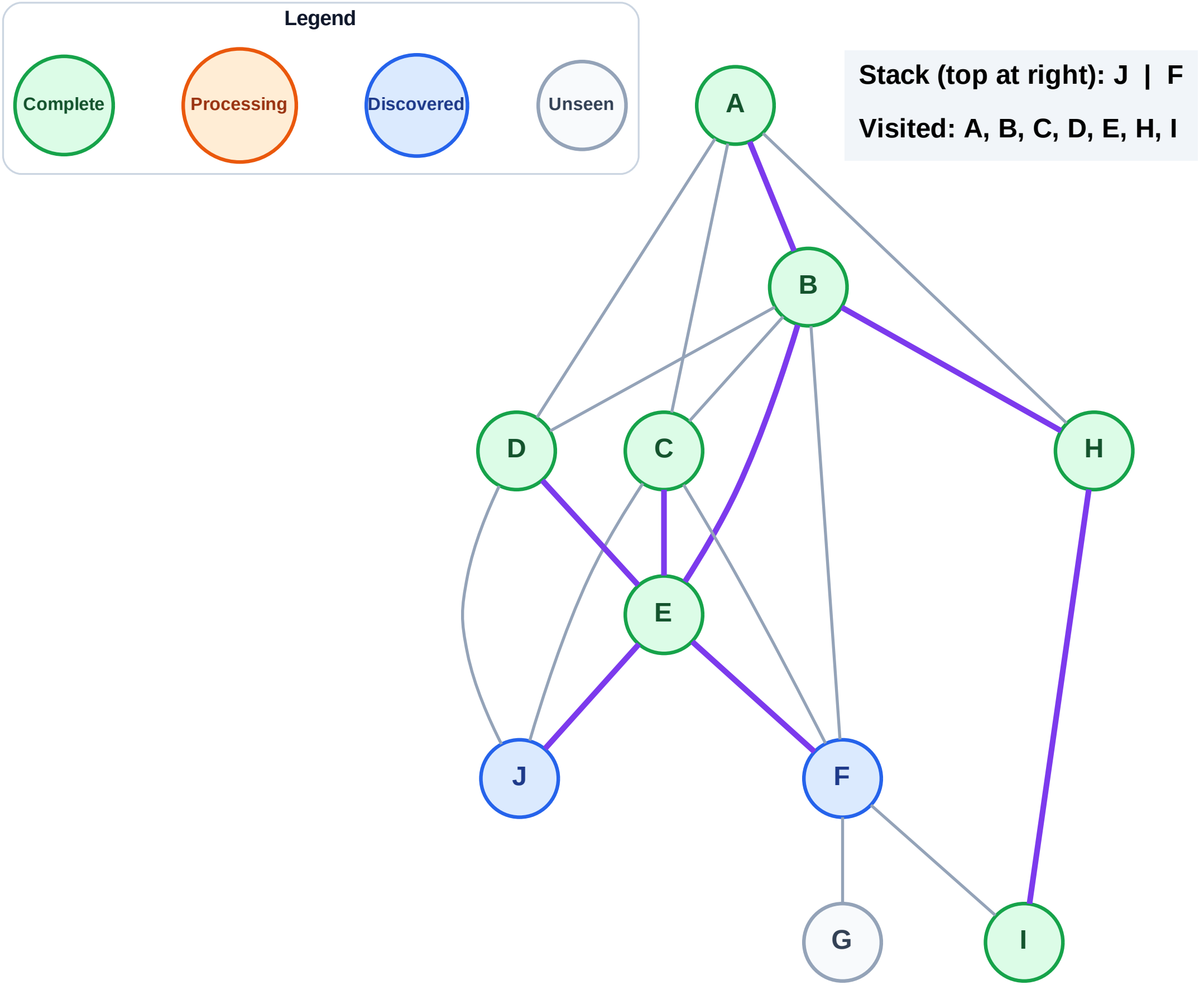
DFS Traversal

Step 14: Process node D



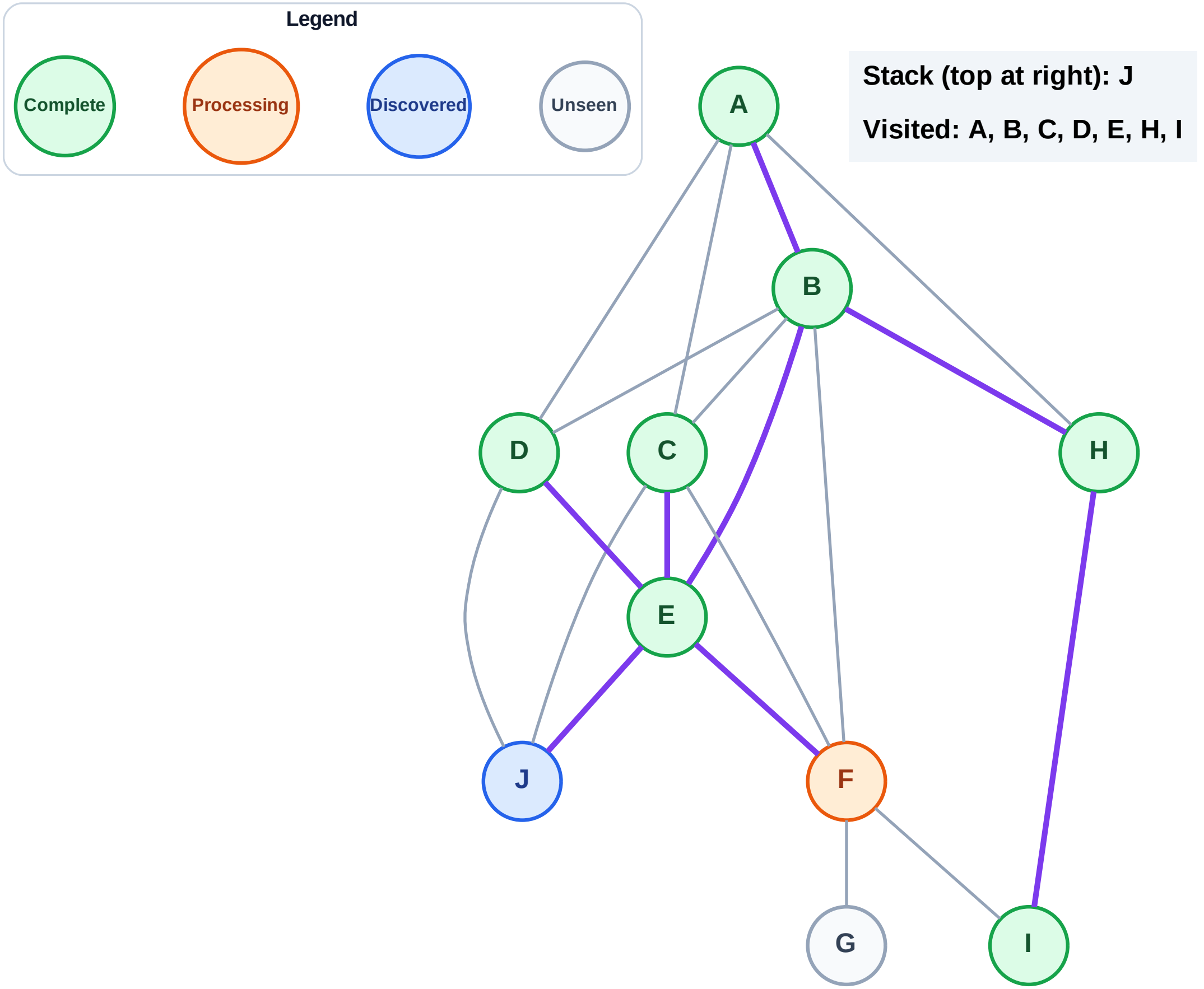
DFS Traversal

Step 15: No new neighbors from D



DFS Traversal

Step 16: Process node F

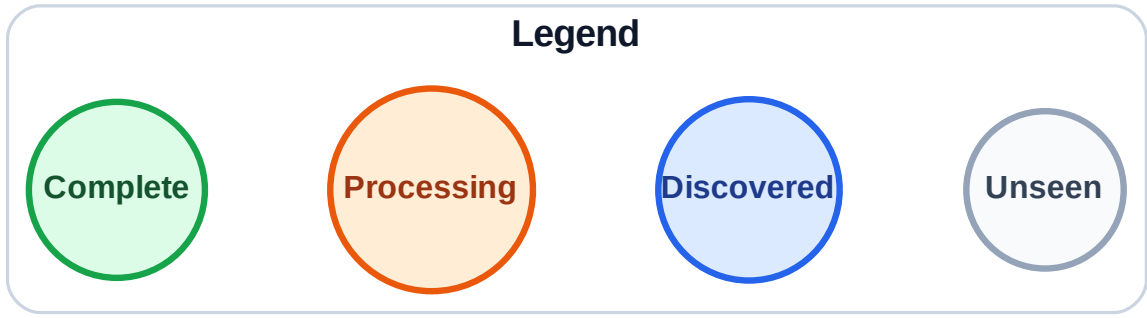


Stack (top at right): J

Visited: A, B, C, D, E, H, I

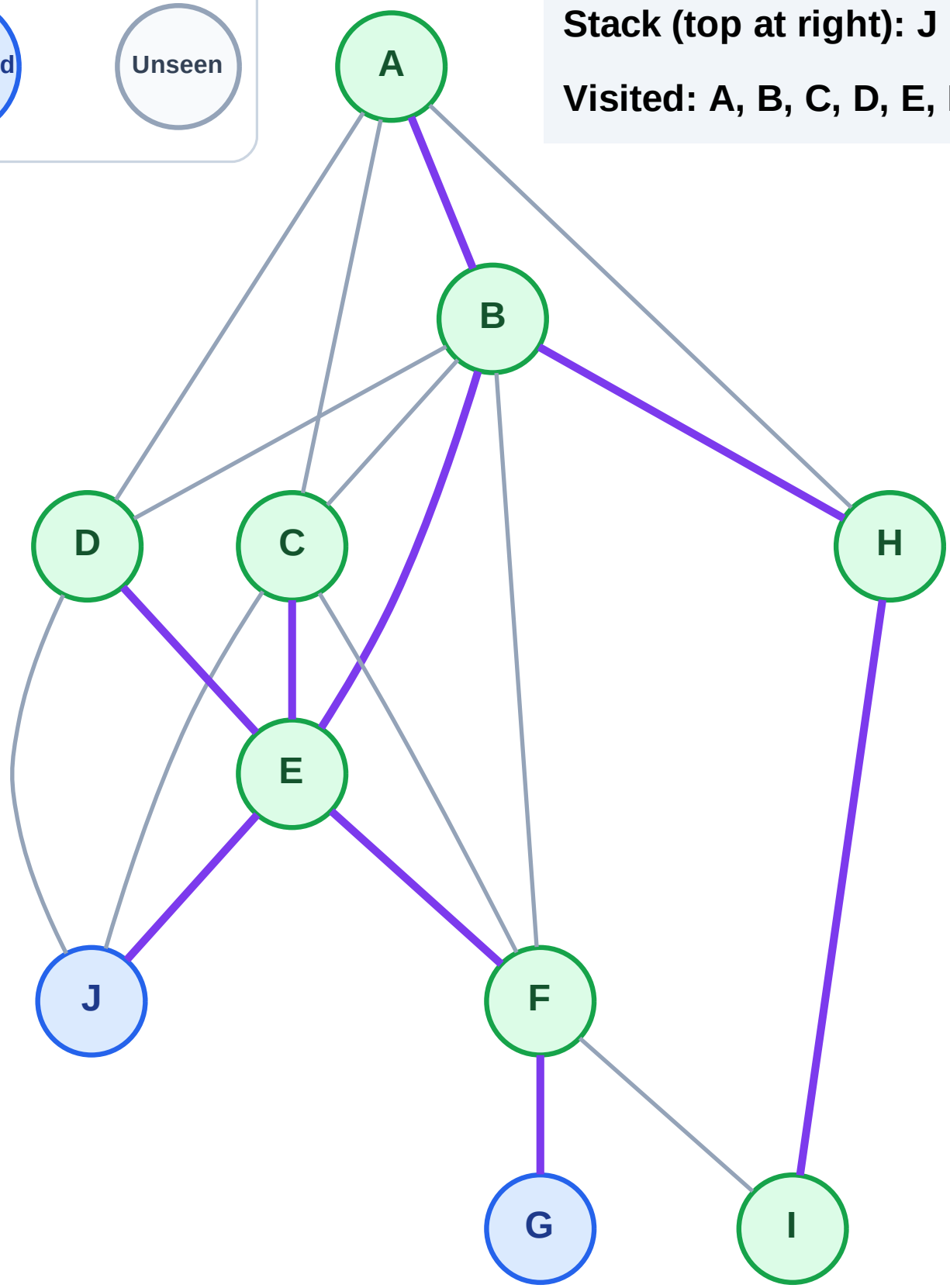
DFS Traversal

Step 17: Discover G from F



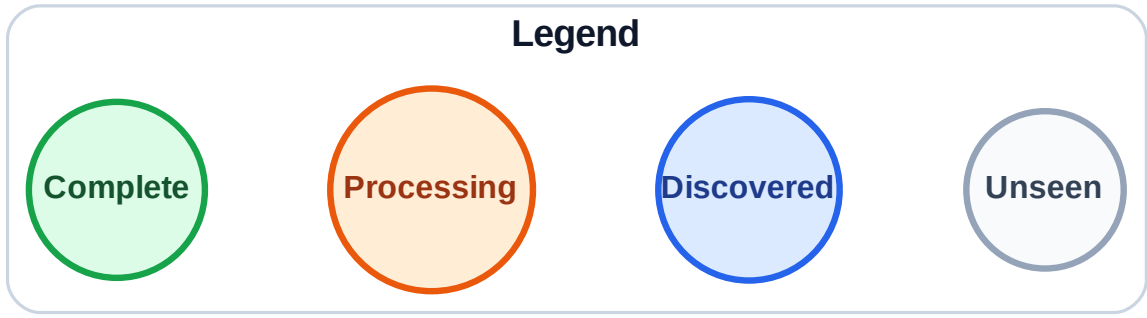
Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



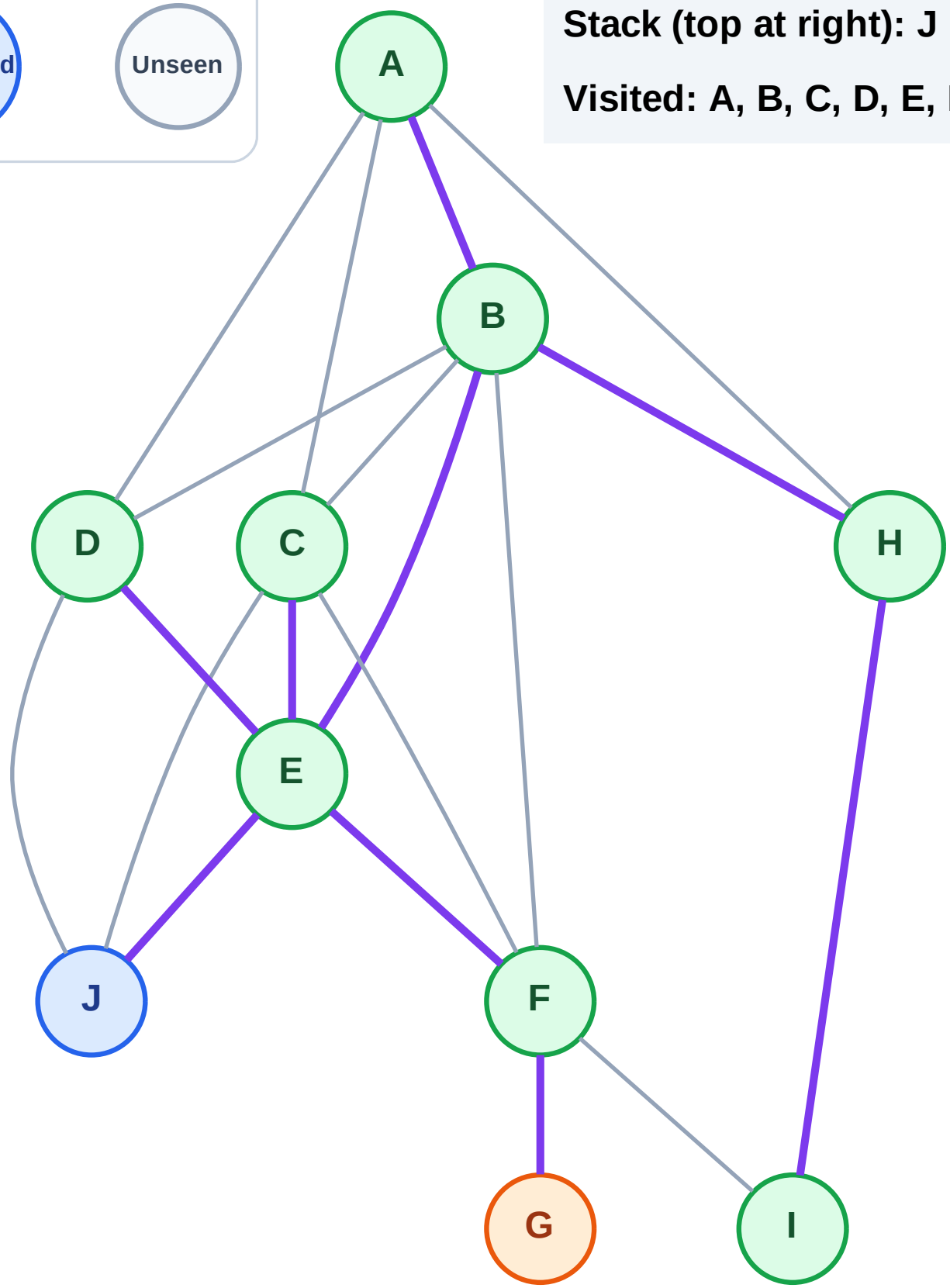
DFS Traversal

Step 18: Process node G



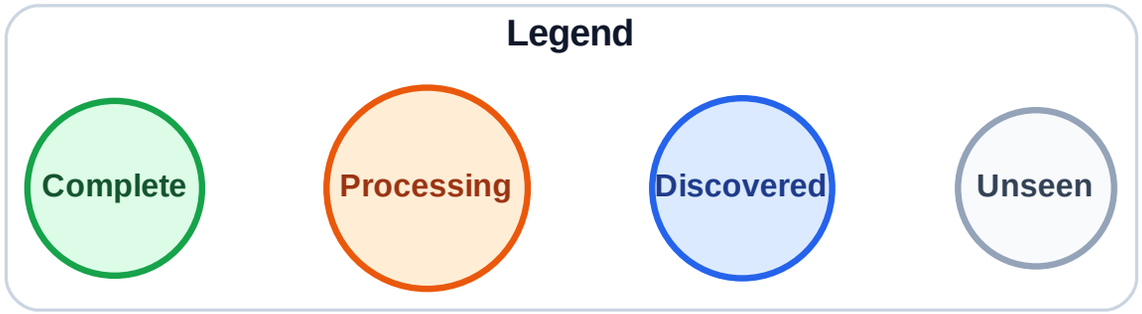
Stack (top at right): J

Visited: A, B, C, D, E, F, H, I



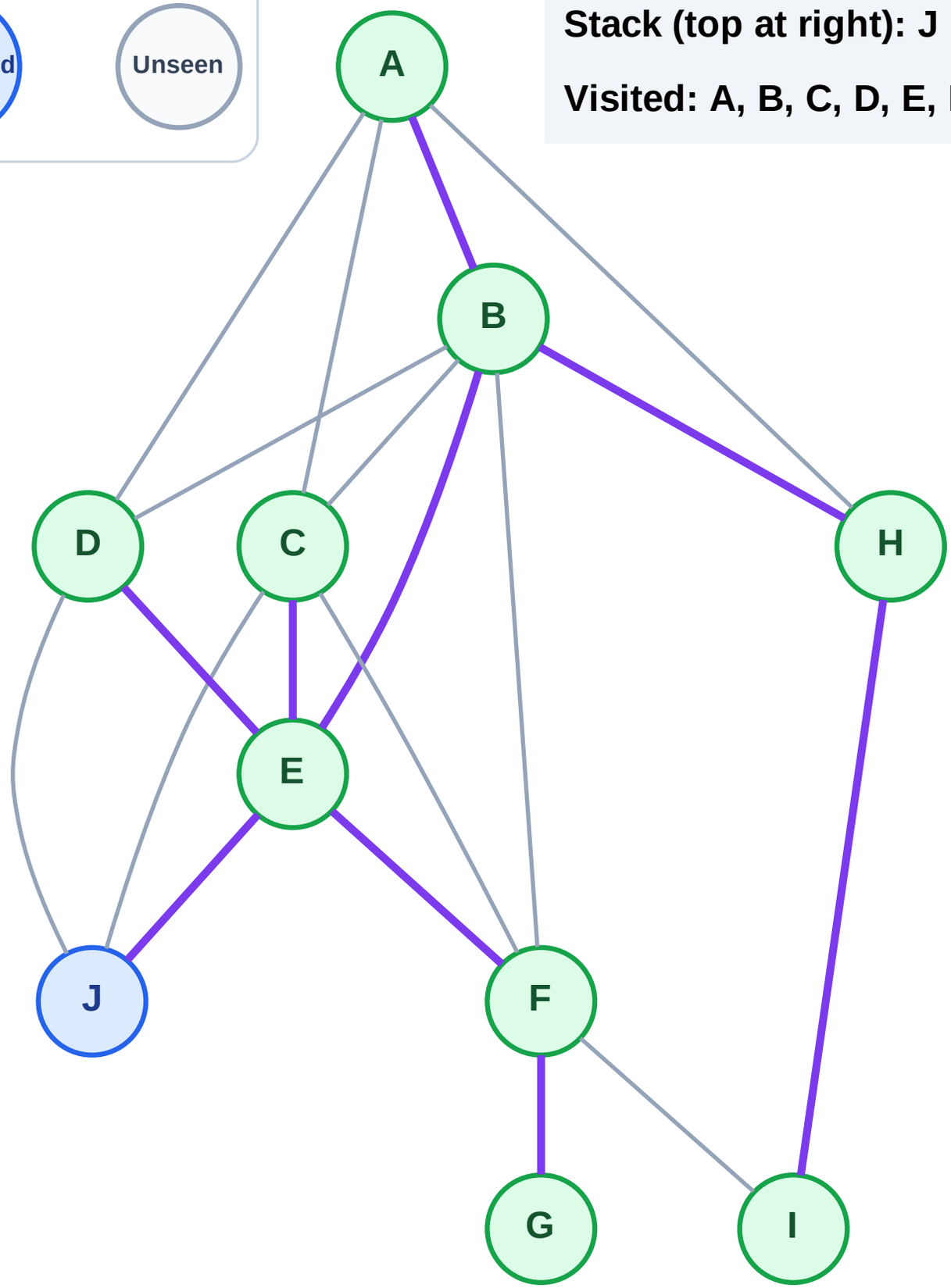
DFS Traversal

Step 19: No new neighbors from G



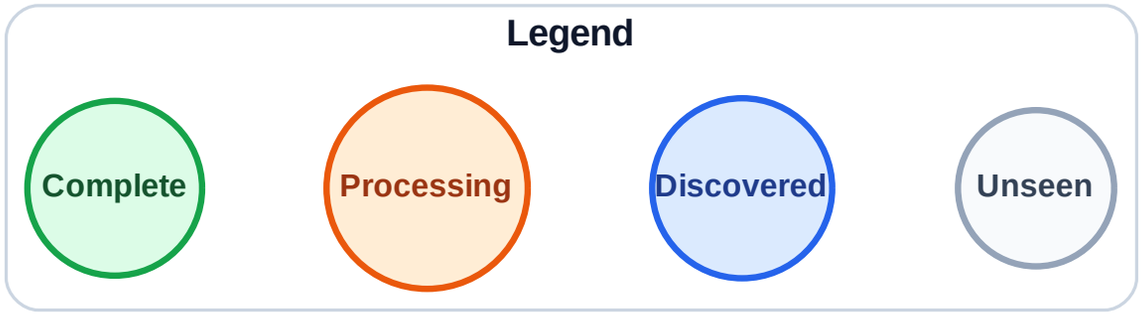
Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



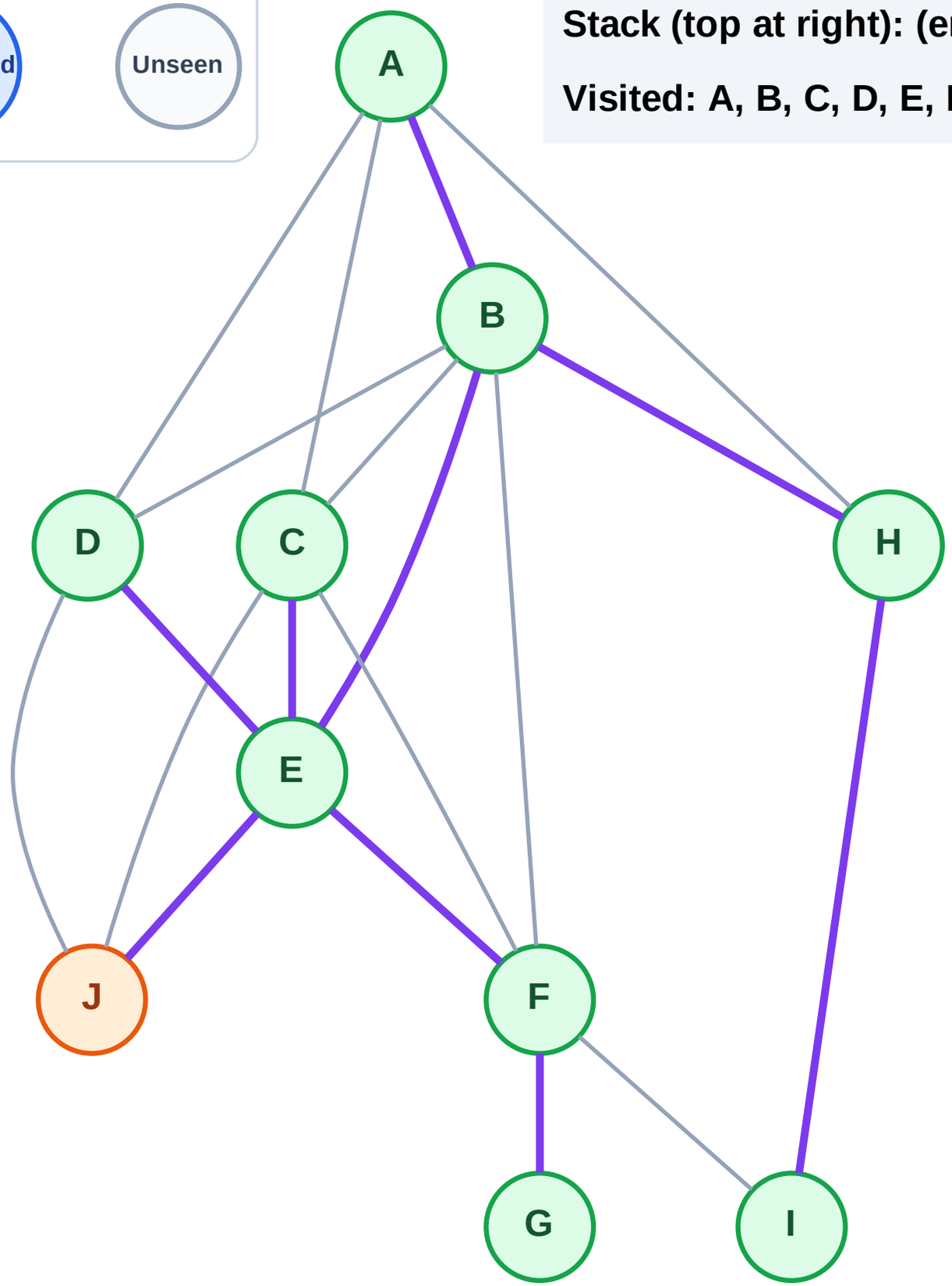
DFS Traversal

Step 20: Process node J



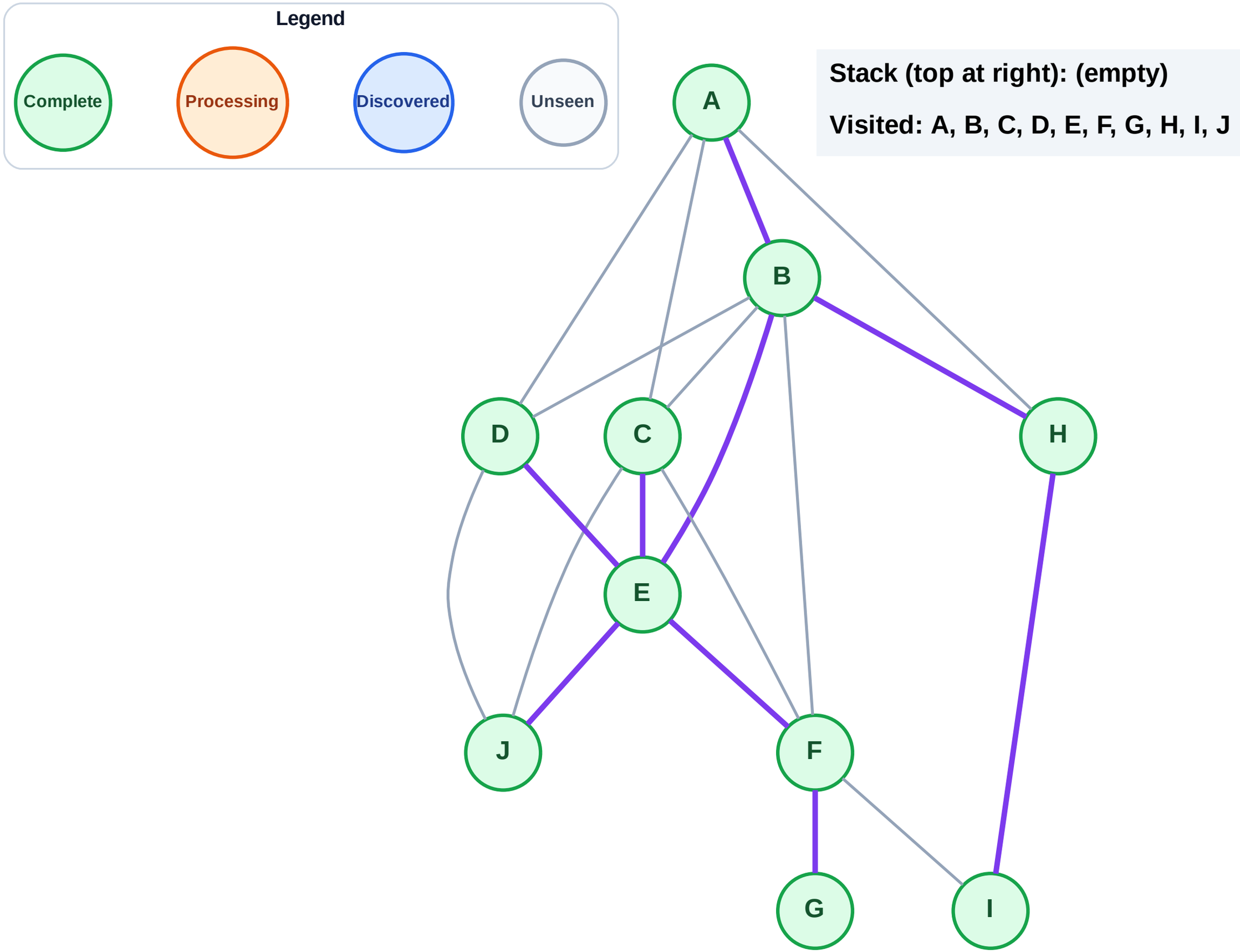
Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J